

A Digital Student Record Management System for Improving Registrar Workflow at Anihan Technical School

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Chapter 1

A Digital Student Record Management System for Improving Registrar Workflow at Anihan Technical School

THE PROBLEM AND ITS BACKGROUND

1.1 Introduction

Anihan Technical School is an all-female Technical and Vocational Education and Training (TVET) institute that has been operating for more than forty years. It was founded by the Foundation for Professional Training, Inc. (FPTI), a non-profit organization that promotes and supports the development of women. The school receives support from partner companies and additional revenue from the sale of its in-house-made “Anihan Goodies” pasulobong products.

The school offers three main courses: Culinary Arts and Restaurant Services (CARS), Watsons Apprenticeship Program (WAP), and Short Courses for Adults. Currently, Anihan only offers CARS to students. Currently, CARS is only provided because there is an insufficient number of students who enrolled for the other courses. Each course is divided into two sections: theoretical and practical. In the theoretical section, students study about 50 subjects over the course of six months. In the practical section, students will do on-the-job training with one of the many industry partners of the school for the course of three months.

The annual student population is approximately composed of 160 students divided into two batches. Each batch does not take the same course section simultaneously—one batch takes the theoretical, while the other takes the practical. Each student is given a one-year scholarship contract that is paid for by a supporter, who is one of the many partner companies. Often, upon graduation, these students would work for one of these companies.

Anihan Technical School is composed of five main departments: Recruitment, Registrar, Faculty, Admin, and On-the-Job (OJT) Office. The Recruitment is responsible for promoting, recruiting, and admitting students to Anihan. The Registrar is responsible for managing school and student documents, as

well as filing Technical Education and Skills Development Authority (TESDA) Special Orders (SO).¹ The Faculty is composed of trainers who teach the theoretical subjects of the courses to students. The Admin manages school operations. And the OJT Office manages industry partnerships and is tasked with securing internship placements for students through these connections.

The primary purpose of this research is to develop a system that would help Aniham Technical School improve the storage, retrieval, and integrity of student records. Additionally, this system aims to prevent the increasing challenge posed by the accumulation of student records as the school continues to operate.

1.2 Statement of the Problem

Aniham Technical School is currently experiencing significant operational inefficiencies and compliance risks due to the accumulation of over 40 years of physical student records. The institution is facing backlogs in complying with TESDA regulatory requirements for SOs, creating delays for approximately sixty students across three batches from 2024. These issues are driven by the reliance on a single staff member to manage information that is scattered across multiple files² and entirely dependent on physical printing. This lack of a centralized record management system makes data consolidation time-consuming, resulting in processing delays that directly hinder the school's daily operations.

These delays are directly caused by a redundant and labor-intensive data management workflow. While the institution has been using Google Forms and Microsoft Excel, the process remains unoptimized. The registrar is still forced to manually export data, format it, and physically cross-check information against handwritten index cards to verify student information. This reliance on a hybrid system of different digital

¹ Special Orders are numbers given by TESDA that clear students of having taken the respective skills program.

² For each student, the files that are stored are personal information, interview evaluation, PSA birth certificate, scholarship agreement signed by the director, good morale certificate, admission form, grades, learners report, certificate of indigency, senior high school student's permanent record, attendance record, and are stored in a green folder.

tools and manual verification creates a bottleneck that prevents the efficient handling of the current eighty-two enrolled students.

Furthermore, the integrity of historical data is severely compromised. The physical archives, containing decades of student data, are stored in a location vulnerable to water leaks that threaten to destroy permanent records. While requests for historical records of alumni prior to 2009 are infrequent, the current method of physical archiving poses a threat to data integrity. The accumulation of unsecured physical records presents significant data security liability, including risks of unauthorized access, theft, and permanent data loss due to the lack of digital redundancy.

Compounding these issues is the severe overload of the institution's human resources. The registrar's office operates with a single personnel member who is simultaneously tasked with billing, student orientation, scholarship management, and IT infrastructure maintenance such as managing routers and CCTVs. This lack of manpower, combined with the absence of a centralized record-keeping system, has resulted in a single point of failure where the resignation or absence of the registrar would paralyze the school's ability to process student credentials and maintain regulatory compliance.

1.3 Objectives of the Study

The school is experiencing a delay in the processing of student records and documents due to the current handling of files and the reliance on a single staff member. Furthermore, the orders are delayed because the student information is scattered across multiple files, making consolidation time-consuming. This lack of centralized and organized record management leads to processing delays that affect the operations of Anihan.

1.3.1 General Objective

The main objective of the study is to address the slow process in student record processing by centralizing the records of the students, which also helps relieve some of the duties of the registrar and digitalizing the accumulated physical records.

1.3.2 Specific Objectives

The specific objectives are as follows:

- a. To design a centralized workflow that organizes, consolidates, and manages all the required information for student records and TESDA Special Order
- b. To implement a secure digital information management system that enables creating, updating, retrieving, and maintaining student records in a centralized platform.
- c. To evaluate the usability, compatibility, functionality, reliability, security, maintainability, and efficiency of the system based on ISO/IEC 25010 software quality standards.
- d. To conduct a time and motion study to test the registrar in their primary functions, comparing both the current method and the provided system.
- e. To evaluate the impact of the system using the following KPIs: improvement in record processing efficiency, Special Order preparation time, data accuracy, user accessibility, and data security

1.4 Scope and Limitations

The general purpose of this study is to develop a centralized student records management system. The study will document the processes of the current records management system of Aniwan Technical School and model a simple, low-cost information system solution to address the identified problems. Furthermore, the enrollment process concerning the creation of a student record will be examined, although its scope lies outside the implementation of the proposed system.

The data will only be gathered within Anihan Technical School. Specifically, the current student record management system and enrollment system, given they are interrelated systems involved in student records management. The gathered data from these two areas will encompass the processes involved in creating, storing, retrieving, and updating student records, along with the corresponding volume of student records involved in these processes.

This study will take place over the course of about one and a half years, divided into three phases, each about four months long. The initial phase involves the investigation of the business processes and issues in the client, leading to the proposal of a possible information system solution. The second phase will be allocated to the development of the proposed system. The third and final phase will be the implementation and testing of the system in the operational environment of the client. Upon the decision of Anihan Technical School to continue using the system, the researchers will commit to post-implementation support to resolve any possible issues or anomalies found and leave it in a sustainable state that requires no future intervention from the researchers.

1.5 Significance of the Study

The study attempts to centralize the student records of the school, providing easier access to the documents and ensuring data security and integrity. The primary beneficiary would be the registrar, who currently handles all encoding, retrieval, and document preparation manually. The proposed system would reduce workload, minimize errors, and shorten the time for processing, solving the issue of slow processing and overload of registrar duties.

The study would also allow students and graduates to benefit from the faster processing of essential documents such as Special Orders, transcripts of records, and certificates of graduation. Having accurate and readily available records improves their employment opportunities, specifically those applying to partner with companies and take opportunities abroad. Faculty and staff will also gain from the system, improving coordination, reporting, and academic monitoring.

This study will also provide a more secure approach to the data management of their records.

Helping safeguard private information, supporting their compliance with the regular requirements requested, and reducing the risks that come with having physical storage.

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Review of Related Literature

The review of related literature to this study is discussed here. The review covers Information Technology, Technical and Vocational Education Training (TVET), and Student Records Management Systems.

2.1.1 Information Technology

Information Technology is fundamentally the study of solutions and needs that bridge the gap between people, information, and current technology (Said et al., 2021, as cited in Muntaka et al., 2024). IT professionals must possess the knowledge, tools, and skills necessary to collaborate effectively with stakeholders and provide appropriate solutions to complex problems (Muntaka et al., 2024). Notably, information technology is recognized as an independent discipline, entirely different from related fields such as information science, computer science, and computer engineering (Basty et al., 2023, as cited in Muntaka et al., 2024).

Information Technology encompasses the use of electronic systems and interactive communication technologies that enable organizations to exchange and manage information efficiently (ScienceDirect Topics, 2024). IT includes a comprehensive range of components such as computers—from mainframes to workstations and desktop personal computers—multimedia devices, telecommunications systems, networks, document reproduction, Artificial Intelligence (AI), and expert systems for speech recognition and knowledge representation (ScienceDirect Topics, 2024). The electronic computer represents only one component within an elaborate and highly differentiated infrastructure that has evolved through successive generations of computing technology.

The scope of IT extends to significantly transforming industries and driving innovation to meet client demands (Muntaka et al., 2024). Understanding IT's development phases helps governments and businesses

worldwide recognize its importance in shaping workforce development, education, and technological integration across diverse sectors (Muntaka et al., 2024).

2.1.2 Technical and Vocational Education Training (TVET)

The quality of the industry depends on the skills and education of its workforce. Technical and Vocational Education and Training (TVET), also known as skill-oriented education, offers the key to technical advancement, rapid industrialization, wealth generation, and poverty reduction (Amin et. al., 2023). As such, many countries worldwide rely on TVET to meet the growing demand for a skilled and competent workforce. TVET focuses on educating job-specific skills as it enhances the employability of individuals in the specific industry they wish to apply for. Through the program, it becomes a tool for sustainable development that increases economic growth, productivity, and human development within the country (Amin et. al., 2023).

In the Philippines, the Technical Education and Skills Development Authority (TESDA) is responsible for the certification of TVET programs employed by various colleges, universities, and institutions. The organization is primarily tasked to manage and supervise technical education and skills development in the country as mandated by Republic Act 7796, also known as the “Technical Education and Skills Development Act of 1994” (Edralin & Pastrana, 2023). To authenticate certificates of graduates of all registered programs, TESDA issues a Special Order (SO), which is a number corresponding to the evaluation of the organization that the graduates have taken the required education and training (TESDA, 2019).

2.1.3 Student Record Management System

Institutions need proper record management to ensure proper documentation of their activities and to make informed management decisions. In the context of an academic setting, management of records includes official documents, books, files, student information, staff, teaching and learning records, and activities. Incorrect data inputs, duplicate data, manual calculations and inputs, and file retrieval are issues associated with academic record management. Given these problems, an effective record management system that can

properly organize, store, and retrieve information is necessary for the efficiency of the institution (Falolo et. al., 2022).

Multiple student records management systems have been used and developed. Previous research has studied these systems.

In a study by Aburime et. al. (2021), the researchers conducted a comparative study of the student records management systems in seven public primary schools of Kwara State, Nigeria. Student documents are physical and are stored in folders, filing cabinets, and shelves. The researchers note three main issues with the current system. First, the documents are prone to deterioration due to the lack of preservation methods. Second, there are little to no security measures to prevent unauthorized access to the files. Lastly, record-keeping was found to be a low priority in these schools, resulting in the falsification of records. Although not explicitly stated, this would imply that document retrieval is difficult and slow. Additionally, the researchers conducted interviews with students and teachers on the importance of a student record management system, with most claiming that it is indeed important and would greatly benefit their study and work.

In a study conducted by Silwamba and Matela (2025), the researchers attempted to develop a digital student record system. The purpose was to address the key challenges of existing student record management systems. These challenges are the growing complexity in managing student records, supporting educational administrators, and encouraging technological advancement. Additionally, the researchers note that an improved system would help foster data-driven decision-making, streamlining administrative work, and creating better parent-school communication. To develop the system, the researchers conducted stakeholder interviews on the student management process and surveys to collect quantitative data on their experiences, satisfaction levels, and expectations of the system. The resulting digital system was three-tiered. The first layer was the presentation layer to directly interact with users. It was responsible for displaying the dashboards, forms, reports, and notifications, while also allowing users to input and view data. The second layer was the application layer, which handles user authentication, record management, report generation, and data validation. The third layer was the data layer that handles the storage of all data. There are three user roles:

student, teacher, and administrator. For security design, password, password encryption, and role-based access control were implemented. Backups of the database were also created to prevent data loss.

A similar study was also conducted by Ramani et. al. (2025). The researchers identified the key needs of students, teachers, and administrators around which the features are designed. The resulting system was built for the main modules: Student Registration, Attendance, Grade Management, and Reports Module. User authentication and database integration were added for security and data storage. For data storage, backups were also created to prevent data loss.

2.1.4 Comparative Analysis of Existing Systems

Table 2.1.4.1 showcases the comparison of features of the different discussed systems.

Table 2.1.4.1 Comparison of Features of Systems

System	User Roles	Data Storage	Data Retrieval	Modular Design	Database	User Authentication
Aburime et. al. (2021)		✓	✓			
Silwamba and Matela (2025)	✓	✓	✓	✓	✓	✓
Ramani et. al. (2025).	✓	✓	✓	✓	✓	✓

In Table 2.1.4.1, the system of Aburime et. al. (2025) is a manual processing system. Compared to the other two systems, it lacks user roles, a modular design, and a database. The system is simply handled by the school's corresponding office, which retrieves and stores documents. Additionally, it lacks modular design and

is simply limited to these two functions. The lack of a proper database is also a result of its current storing method through folders, cabinets, and shelves. The other systems are likewise similar, employing the three different user roles (students, teachers, administrators), data storage and retrieval, modular design—features based on the functions and needs of the users—and the use of a database with backup.

2.2 Conceptual Model of the Study

The conceptual model of the study uses the Input-Process-Output (with feedback) model, which shows the flow of developing the student document management system for Aniham Technical School. Figure 2.2.1 showcases this model in detail.

The input phase includes all the relevant data gathered to understand how Aniham currently manages its student records. This includes physical folders and documents, manual encoding procedures, Google forms, Word documents, Excel spreadsheets, TESDA SO template and requirements, as well as other insites and observations gathered from the on-site visit and interviews.

The process phase uses the collected information to plan, design, program, document, and test the proposed Student Record Management System. These steps are used to help establish a centralized workflow that addresses the problem in their existing method.

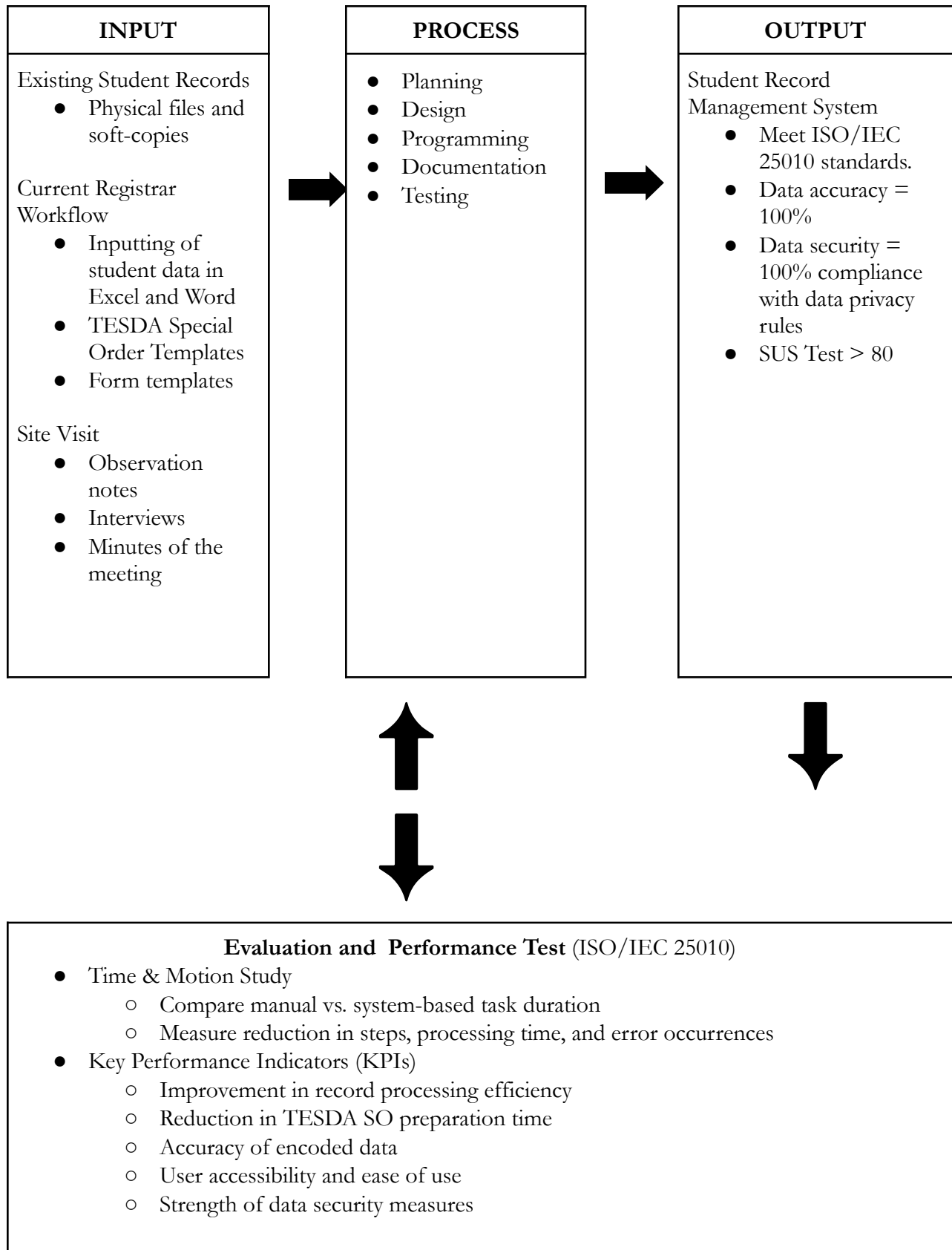


Figure 2.2.1 Conceptual Model of the Study

Lastly, the output phase results in a fully functioning Student Record Management System that centralizes student information, reduces the registrar's workload, and improves processing speed, data accuracy, accessibility, and data security. After deployment, the system will undergo a structured evaluation guided by ISO/IEC 25010 standards to assess its functionality, reliability, usability, efficiency, and maintainability. In addition to this qualitative assessment, the system will be measured through specific performance benchmarks. A Time and Motion Study will compare manual versus system-based task duration, including reductions in steps, processing time, and error occurrences. Furthermore, Key Performance Indicators (KPIs) will be used to evaluate improvements in record processing efficiency, the reduction of TESDA Special Order (SO) preparation time, the accuracy of encoded data, user accessibility and ease of use, and the overall strength of data security measures. The results from these evaluations will guide refinements to ensure the system effectively supports the school's operations and long-term goals.

2.2.3 Digitization Plan

The central objective of the Student Record Management System is to eliminate the reliance on physical records that have accumulated over more than forty years of Anihan Technical School's operation. The digitization plan addresses both the backlog of 120 existing student records and the systematic onboarding of all future student records directly into the SRMS from the point of deployment. Table 2.4.1 details the digitization plan of existing and future student records.

Table 2.2.3.1 Digitization Plan for Existing and Future Student Records

Component	Details
Scope of Digitization	
Target Records	150 existing student records (from batches currently on file at Anihan Technical School, covering records of 2025).
Record Components	Per student: personal information, interview evaluation, PSA birth certificate, scholarship agreement, good morale certificate, admission form, grades, learner's report, certificate of indigency, SHS permanent record, and attendance record — stored in a single digital profile within the SRMS.
Future Records	All incoming student batches (approximately 80 students per batch, two batches per academic year) will be enrolled directly into the SRMS from the point of deployment forward, eliminating new physical filing.
Digitization for Existing Records Workflow	
Step 1 – Preparation	The Registrar retrieves existing physical green folders from the Student Documents Cabinet and organizes them by batch code (e.g., QSRO5, QSRO6, CARS1) and section.
Step 2 – Student Record Creation	For each student, the Admin or Registrar creates a new StudentRecord in the SRMS, inputting all personal information fields as defined in the Data Dictionary (Table 3.2.8.1).
Step 3 – Document Scanning & Upload	Physical documents (PSA, Form 137, Form 138, admission form, etc.) are scanned and uploaded via the encodeStudentDocsPerSection() or encodeStudentDocsPerBatch() function. Files are stored as BLOBs in the documents table.
Step 4 – Grade Encoding	Historical grades from existing Excel spreadsheets and printed grade sheets are entered into the grades table, linked to the respective student_id and subject_code.
Step 5 – Verification	The Registrar cross-checks the encoded digital records against the physical source documents to ensure 100% data accuracy, as per the KPI requirement.
Step 6 – Status Update	Upon completion of a student's record, the student_status field is updated to either 'Active' or 'Graduated' to properly classify current and alumni records.

The researchers will facilitate the digitization of the existing student records with the assistance of staff from the Office of the Registrar. To expedite and facilitate computer skills training, students from the

school may assist as well. On scheduled onsite visits, the team aims to digitize fifteen student records and related documents per visit. The digitization will take place strictly within the school premises, and no records or documents should be brought outside for data security reasons. This work setup allows the researchers to test the system and train staff in its use.

In the case of the future records, the researchers must have the admin, registrar, and student user role functionalities (to be discussed in 3.2.9) in time for the enrollment period of Anihan Technical School. During that period, the students will simply input their personal data to create the records.. Management of the records will be handled by the staff, with the assistance of the researchers, should the need arise. However, the overall finished system must handle all future records creation as set by the objectives

2.3 Assessment of Alternatives

Multiple options are available to resolve the issues of Anihan Technical School. Based on the analysis of the researchers, there are five alternatives that could be considered. Each was assessed through a decision matrix (see Table 2.3.1). The assessment of alternatives was based on the following criteria: Cost, Ease of Data Retrieval, Ease of Data Encoding, Storage, and Data Security. These were considered based on the needs and issues of the client (see Chapter 3, 3.1 Systems Analysis for further details). Ease of Data Encoding was assigned the highest weight of five to prioritize solutions that minimize the workload of the registrar. Ease of Data Retrieval followed with a weight of four, reflecting the urgent need to locate historical records for pending Special Orders. Storage was assigned a weight of three to accommodate the digitization of the current and future student records. Cost was assigned a weight of two, indicating that the school is willing to invest in a system that resolves the current operational inefficiencies. Finally, Data Security is weighted at one, focusing on internal data integrity rather than external threats.

Table 2.3.1 Decision Matrix of Alternatives

Alternatives	Cost (2)	Ease of Data Retrieval (4)	Ease of Data Encoding (5)	Storage (3)	Data Security (1)	Score
Current System	5	3	1	3	1	37
Hire More Employees	3	4	3	3	2	48
In-house Server Database	2	5	5	4	5	66
Cloud Server Database	3	5	5	3	4	64
Open Source Student Information System (OpenSIS)	4	4	3	4	3	54

2.3.1 Current System

This is the current workflow of the school, which utilizes a combination of different digital tools: Microsoft Excel for data storage, Microsoft Word for document creation, and Google Forms for initial student data collection.

In the evaluation, this alternative received a total score of 37. It achieved a high rating in Cost because it utilizes software the school already possesses. However, it was given a one in Ease of Data Encoding because of the slow process it takes to encode the student's information. Although data is digital, the workflow is fragmented; the sole registrar must manually export data from Google Forms, reformat it for Excel, and copy-paste details into Word for reports. This lack of integration creates a heavy workload for the already overloaded personnel. Storage is rated at three because while files are digital, they are stored in local folders without centralized database structure, making them prone to version conflicts or accidental deletion. Data Security also received the lowest rating of one because Excel and Word files lack role-based access controls and audit trails, making the data vulnerable to unauthorized copying or modification. Furthermore,

they are still utilizing physical records and all files are stored in the registrar, which anyone that is a personnel other than the registrar could enter, which produces the risk of theft.

2.3.2 Hire More Employees

This alternative entails hiring additional personnel for the registrar to assist with encoding and file management. This alternative resulted in a total score of forty-eight. While adding manpower improves Ease of Data Retrieval by allowing more people to search through the Excel files and folders, it does not solve the underlying issue of fragmented tools. Ease of Data Encoding remains a score of three as the process of storing, encoding, and retrieving still relies on Word and Excel. Now, the cost has a score of three and is lower than the current system due to the recurring expense of salaries. Data Security remains low, as increasing the number of personnel with access to the student data stored in Excel files increases the risk of data mishandling.

2.3.3 In-house Server Database

This alternative proposes a custom web application hosted on a physical server within the school premises, operating on a Local Area Network (LAN). This solution has a total score of sixty-six. It achieved perfect scores for Ease of Data Retrieval and Ease of Data Encoding because the web application centralizes the functions of Google Forms, Excel, and Word into one platform. The registrar can generate reports and search records instantly without switching to different applications. The Storage has a score of four and Data Security with a score of five because the data is stored on a dedicated local server with strict user access controls, preventing the issues found in the current file-based system. Although it has a score in Cost of only two due to hardware procurement, the efficiency gains outweigh this disadvantage.

2.3.4 Cloud Server Database

This involves a custom web-based application hosted on a remote cloud server, accessible via the internet. This alternative has a total score of sixty-four. While it matches the in-house alternative in retrieval and encoding efficiency, it has a lower score in Storage, with a score of three, and Data Security, with a score of

four. This is because cloud storage often comes with tiered pricing limits, and the security of the data is partially dependent on third-party providers and internet stability. It did score better in Cost with a score of three as it avoids the upfront hardware expense of a physical server.

2.3.5 Open Source Student Information (OpenSIS)

This option suggests adopting a generic, free-to-use software solution like OpenSIS. This option has a total score of 54. While it is cost-effective, it scored lower in Ease of Data Encoding with a score of three compared to custom solutions. Generic systems often require complex configuration to match the specific “Special Order” formats of TESDA, potentially requiring the registrar to revert to Excel for certain tasks. Storage is rated at four as it utilizes local hardware, but Data Security is rated at three due to the reliance on public codebases that may require specialized maintenance the school cannot currently provide.

Based on the decision matrix, the researchers have selected the In-house Server Database as the best alternative. This solution resolves the inefficiency of the current system by consolidating the functions of Excel, Word, and Google Forms into a single dedicated web application. This approach supports the sole registrar by providing a unified platform where student information can be retrieved and utilized for reports efficiently, effectively eliminating the need for manual reformatting and redundant data transfer between different files.

Chapter 3

RESEARCH METHODOLOGY AND TECHNICAL BACKGROUND

3.1 System Analysis

There are two main systems that are involved concerning student records in Anihan Technical School: The Enrollment System and the Student Records Management System. The enrollment system is the first in the workflow of the client, receiving the student information and documents to create the student record. While the student record management System is the one that manages the student records by encoding and retrieving them.

3.1.1 Data Flow Diagram

To discuss the two systems in the workflow of student records management, the first system will be discussed first. Figure 3.1.1.1 is a context-level Data Flow Diagram (DFD) that illustrates the flow of data between entities and the current enrollment system of Anihan Technical School.

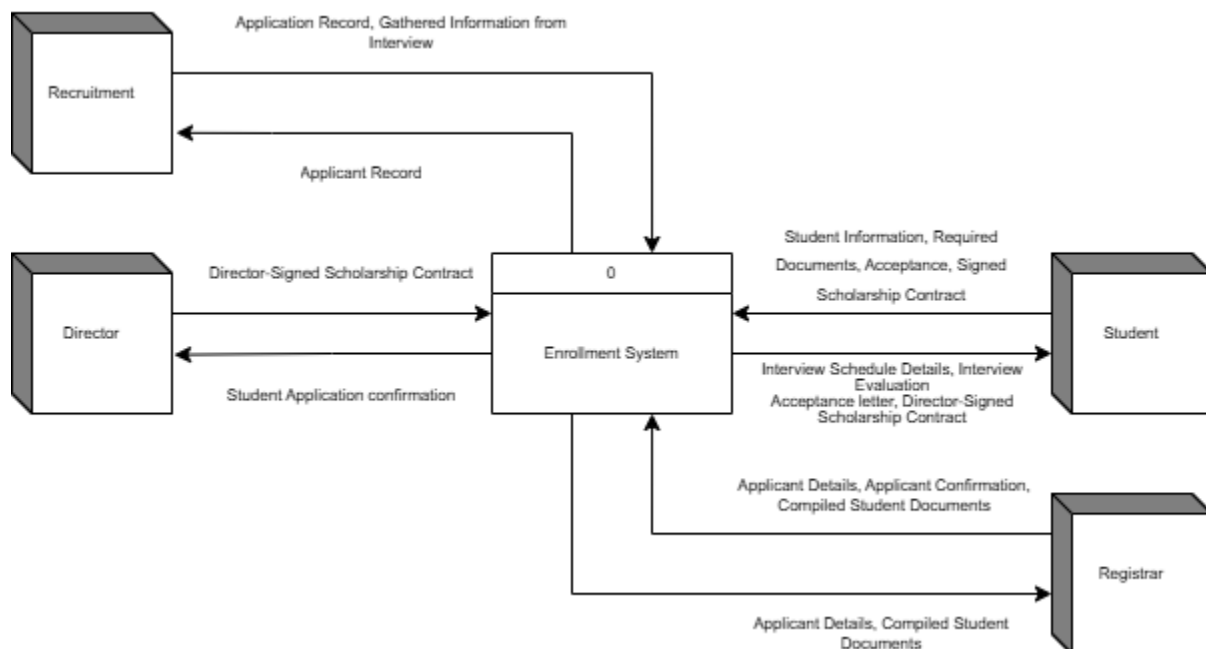


Figure 3.1.1.1 Enrollment System Context Level

Four entities interact with the enrollment system: Student, Registrar, Director, and Recruitment. For the first entity, students input their student information, the required documents for enrollment, and the signed scholarship contract. The system would send them the interview schedule details, interview evaluation, the acceptance letter, and the director-signed scholarship contract. The second entity, the registrar, receives the applicant details of the student and the compiled student documents. In return, they forward the applicant details and compiled student documents, and input the application confirmation. For the third entity, the director receives the student application confirmation and returns the director-signed scholarship contract. The last entity in the workflow is the recruitment, which receives the student details and returns the interview schedule details and the gathered information from the interview.

Figure 3.1.1.2 is the level 0 diagram of the enrollment process. It illustrates the exact data flows and transformations as they are passed between entities and data stores.

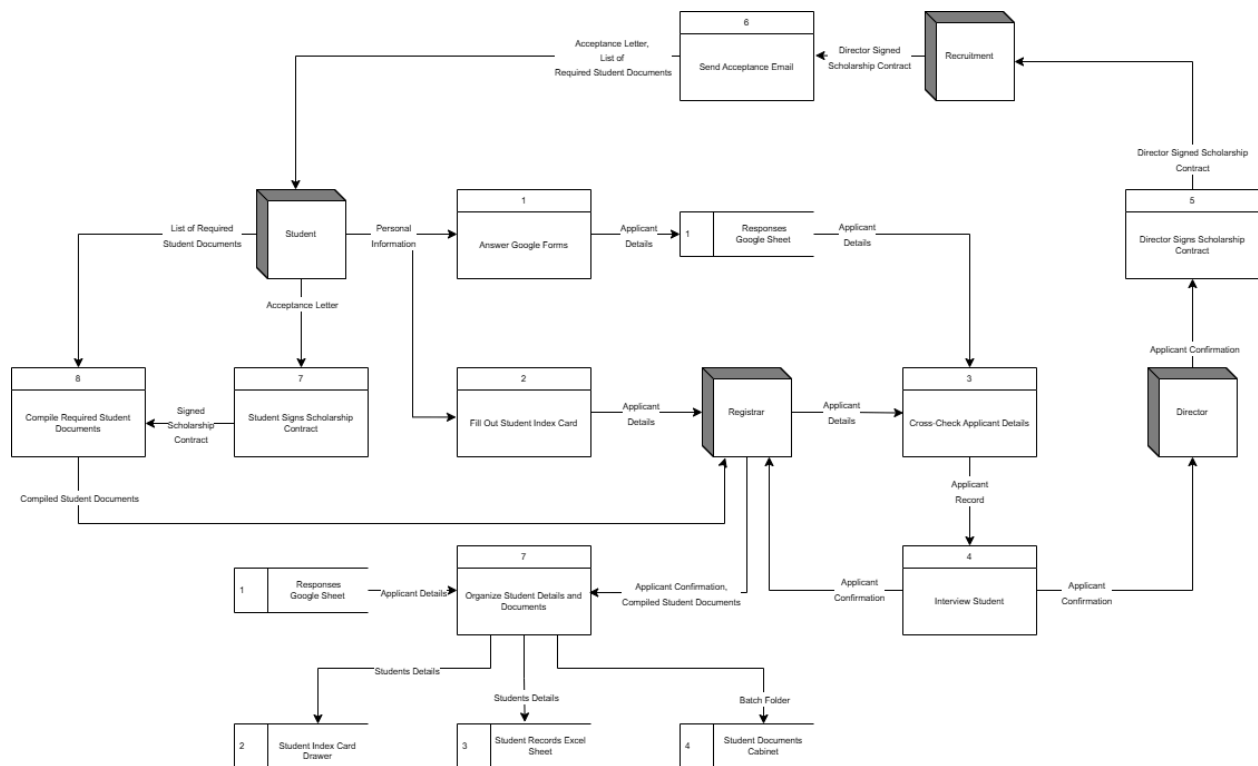


Figure 3.1.1.2 Enrollment System Level 0

At the start of the process, the student is an applicant who provides their personal information through the application Google Form and the student index card. The applicant's details response from the Google Form will be stored in the Responses Google Sheet, while the applicant's details from the student index card are passed to the registrar. The registrar cross-checks the content from both sources and then forwards it for the student interview. The interview process returns the confirmation of the applicant to the director, who then signs the scholarship contract. This document is forwarded to the recruitment department and is attached to the acceptance letter, alongside the list of required student documents that the student must submit, and is sent to the student. The documents required for submission are a certificate of indigency, Form 137, Form 138, and ID pictures. The student receives it, signs the contract, and submits it with the compiled student documents to the registrar. The registrar finally stores the documents and student details in the Student Index Card Drawer, Student Records Excel Sheet, and Student Documents Cabinet, while using the details from the Responses Excel Sheet and applicant confirmation for reference.

Currently, the method of storing data in the last three data stores of the process is inefficient. Each student index card stored within the drawer functions as the physical student index card. Each of them is grouped by batch. For the student documents, each student has their own folder stored in the cabinet. Other student documents gathered throughout the year are stored in their respective folders and batch folders. The data store Student Records Excel Sheet, which was previously discussed, is a recent attempt to digitize the student records. Each batch has its respective sheet, with each row pertaining to a student record alongside their grades and attendance. In both physical data stores, the organization of documents and records is only known by the registrar. While for the digital data store, each sheet is stored in different folders within the desktop of registrar. Backups of these files are also stored in external hard drives, but are at risk of being damaged, making them inoperable.

For the interview process, it is an explosion level of the enrollment system. Figure 3.1.1.3 illustrates the process.

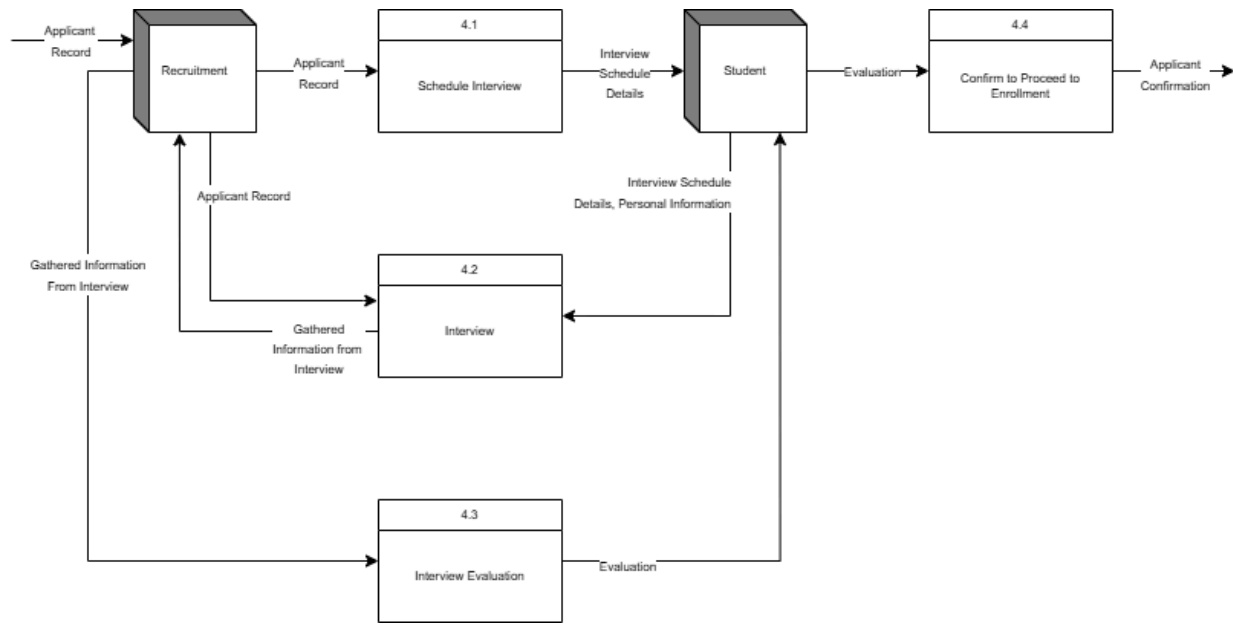


Figure 3.1.1.3 Interview Process Explosion Level

In this process, the recruitment receives the applicant record and schedules an interview with the student. After the interview, the recruiter evaluates the student based on the gathered information. Once the evaluation is done, they present it to the student, and the student provides their confirmation, should they wish to continue with the enrollment.

After the student is finally enrolled, the Student Records Management System takes place. This process operates throughout most of the academic year. Figure 3.1.1.4 is a context-level diagram of this system.

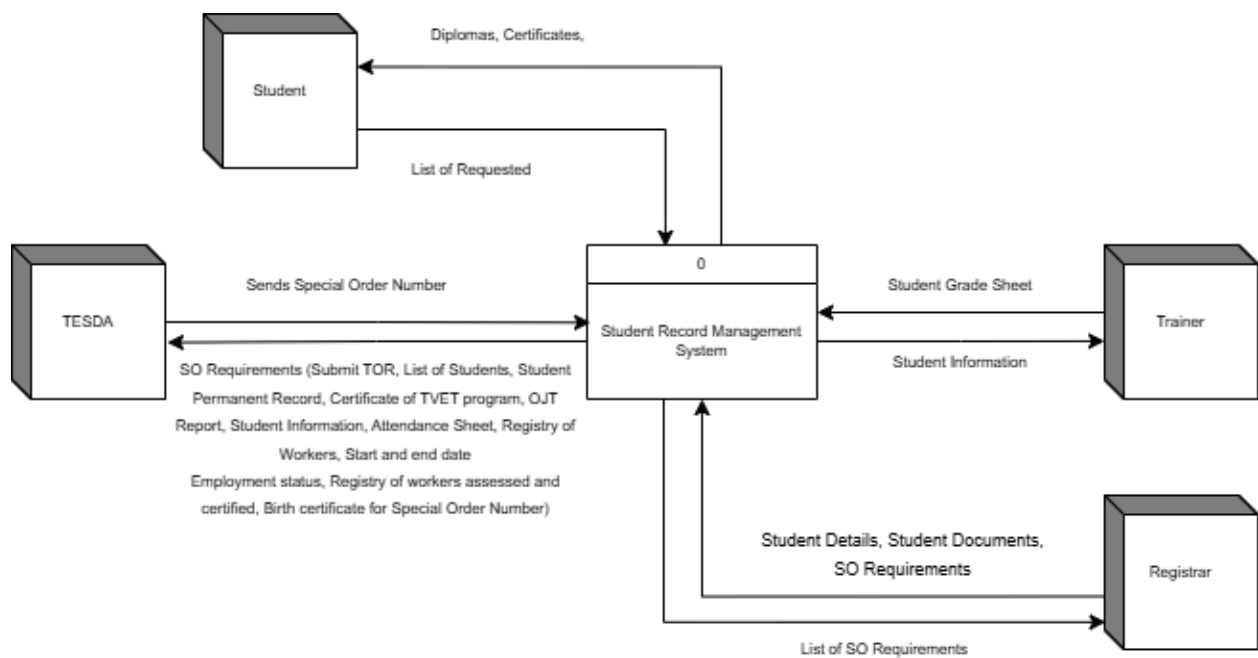


Figure 3.1.1.4 Student Records Management System Context Level

Four entities are interacting with the student records management system: Registrar, Trainer, TESDA, and Student. The Registrar inputs the student details, student documents, and SO Requirements—which the list of SO requirements comes from TESDA. Trainers provide the student grade sheet, referencing the respective student information of the student to be graded. TESDA receives the SO requirements (TOR, List of Students, Student Permanent Record, Certificate of TVET Program, OJT Report, Student Information, Attendance Sheet, Registry of Workers, Start and End Date, Employment Status, Registry of Workers Assessed, PSA Birth Certificate) and then sends the SO number. The system provides the student with their diploma and certificates. The student could also request documents and receive them.

Figure 3.1.1.5 is the level 0 diagram of the student records management process. It illustrates the exact data flows and transformations as they are passed between entities and data stores.

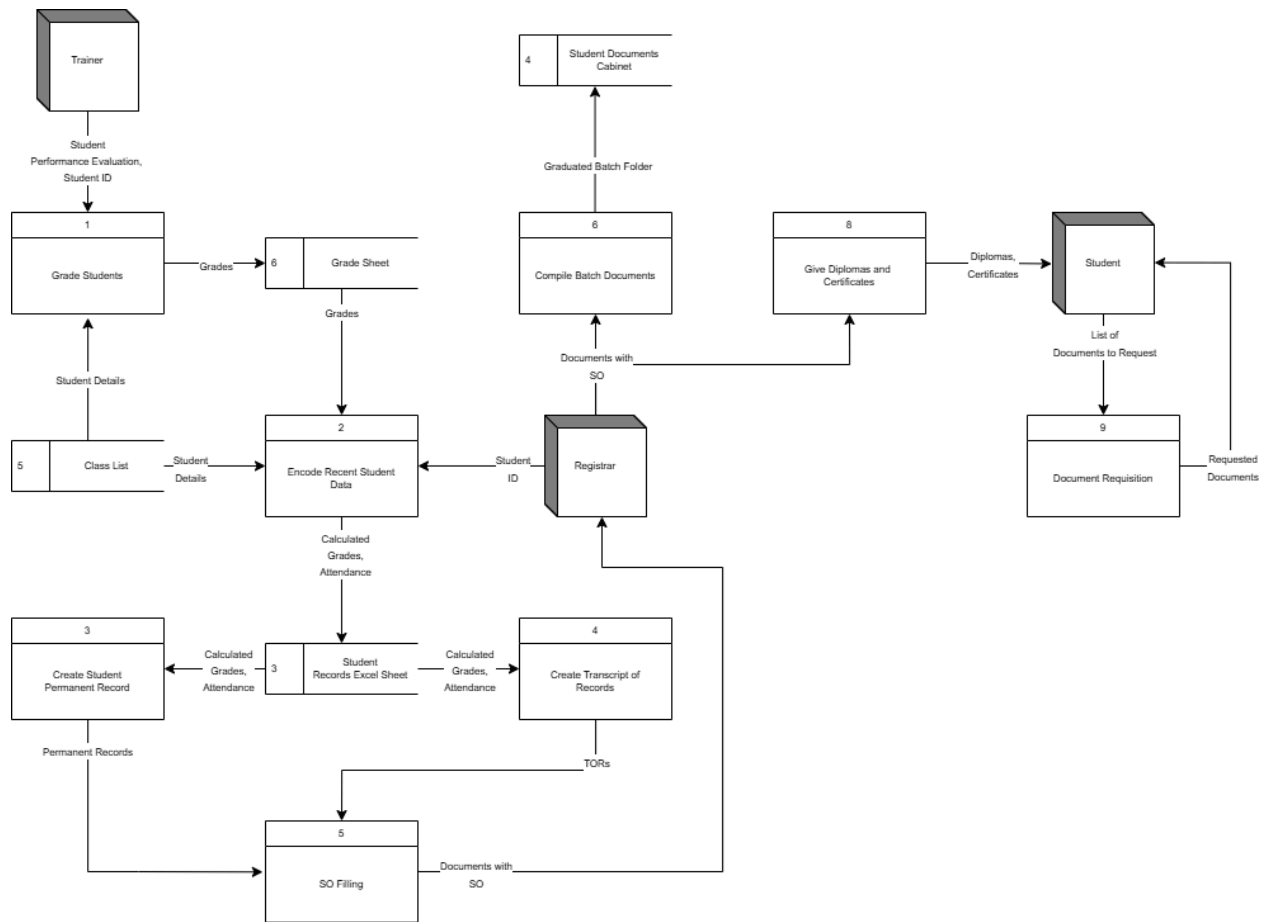


Figure 3.1.1.5 Student Records Management System Level 0

The process starts with the trainer submitting the grades of the student, recorded in the grade sheet, to the registrar. Using the student ID, the registrar encodes the grades in the Student Records Excel Sheet. From this sheet, they use the calculated grades and attendance of the student to create the student's permanent record and TOR. These two documents will be compiled with other documents in the SO filing process. Once the SO is filed, the registrar receives the documents with the SO number and compiles the batch documents to be stored in the Student Documents Cabinet. The diplomas and certificates, now with the SO number, are then provided to the student after one month of the student's graduation. Conversely, the student could also request documents from the registrar during and after their stay in Anihan Technical School.

Currently, the registrar is using multiple documents to format and store the student's records—student grades, grade sheet of each trainers, transcript of records, student permanent record—this is a slow and

cumbersome process as it requires the personnel to go through multiple documents and copy and paste the information to each documents, this is prone to errors so this will require checking of each information. This impacts the process of requesting the special order number from TESDA, as there is a lot of data and information that needs to be formatted and submitted to TESDA in order for the students to get their diplomas, transcripts of records, and certificates of competency. Currently, there are 3 batches of students who are delayed due to this problem.

There are two explosion levels in this system: SO Filing and Document Requisition. They are illustrated in Figures 3.1.1.6 and 3.1.1.7.

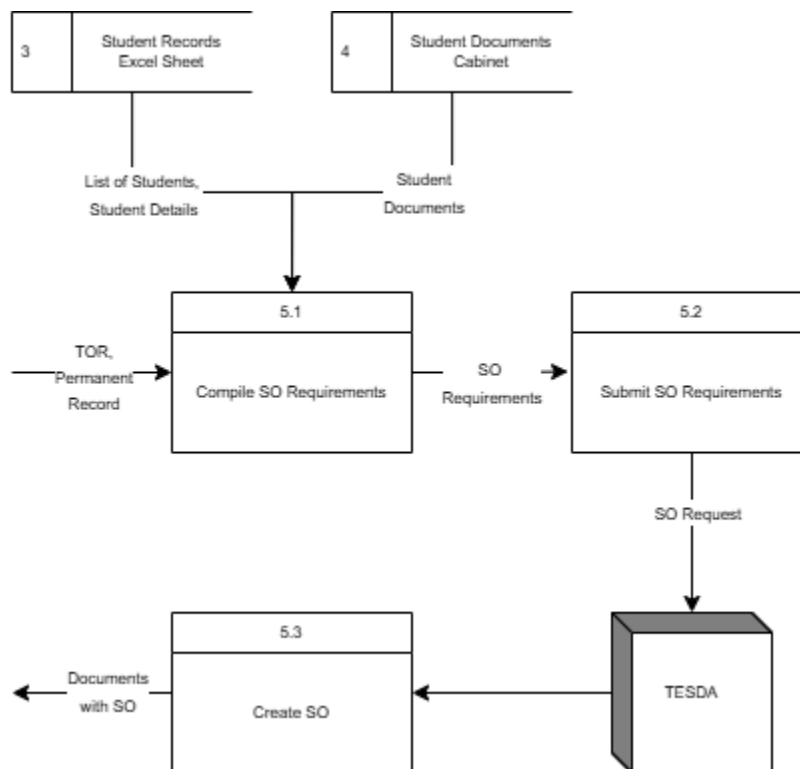


Figure 3.1.1.6 SO Filing Explosion Level

At the SO Filing explosion level, the Registrar compiles the SO requirements, taking the TOR, permanent record, list of students, their details, and other corresponding student documents. The SO requirements are submitted to the TESDA, which the TESDA takes the request and creates an SO number.

The SO number is stamped on the submitted documents and returned to the registrar. It should be noted that TESDA takes about five months to create an SO number and return the documents.

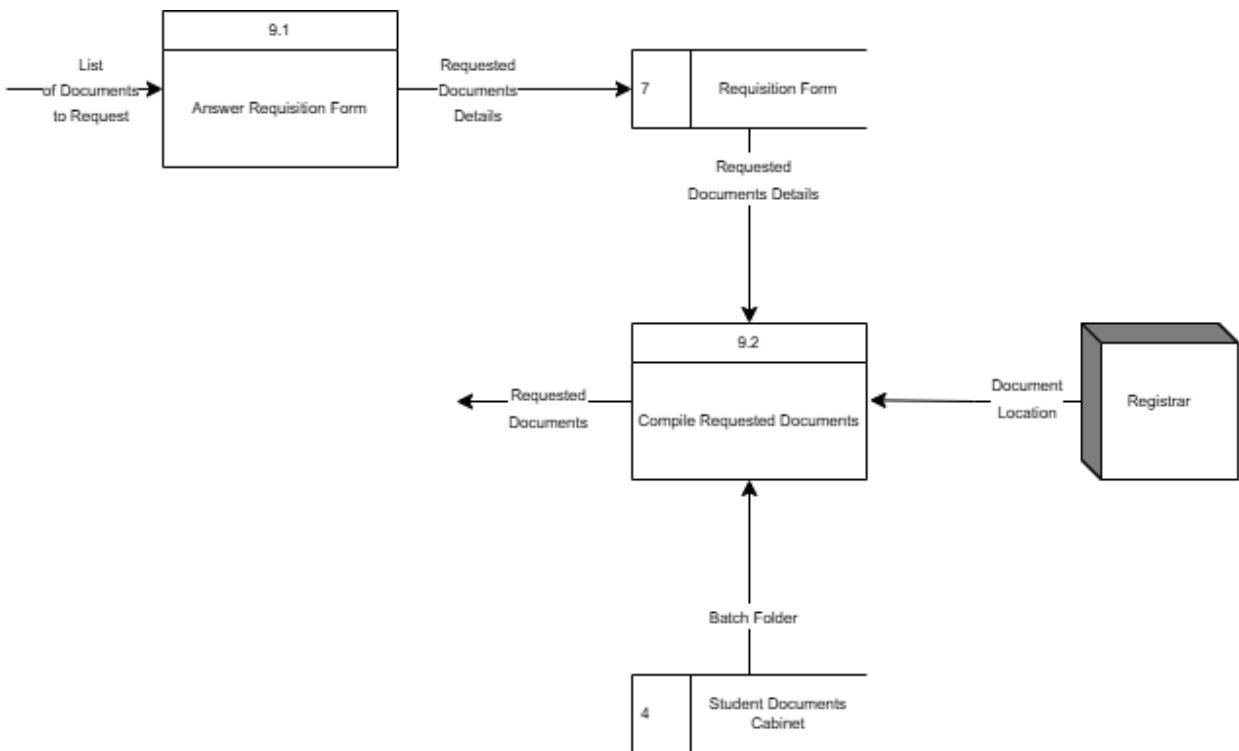


Figure 3.1.1.7 Document Requisition Explosion Level

At the Document Requisition Explosion Level, the student could submit a list of documents to request from the registrar by filing a requisition form. The registrar gets the details of the requested documents from the form and compiles the requested documents, based on the corresponding batch folder in the Student Documents Cabinet. The documents are then given to the student.

3.1.2 Business Process Modelling and Notation

The business process workflow of the two systems is illustrated through Business Process Modelling and Notation (BPMN) diagrams. These diagrams describe the actions and decisions that each entity performs. Figures 3.1.2.1 and 3.1.2.2 are the BPMN diagrams of the enrollment system and student record management system, respectively.

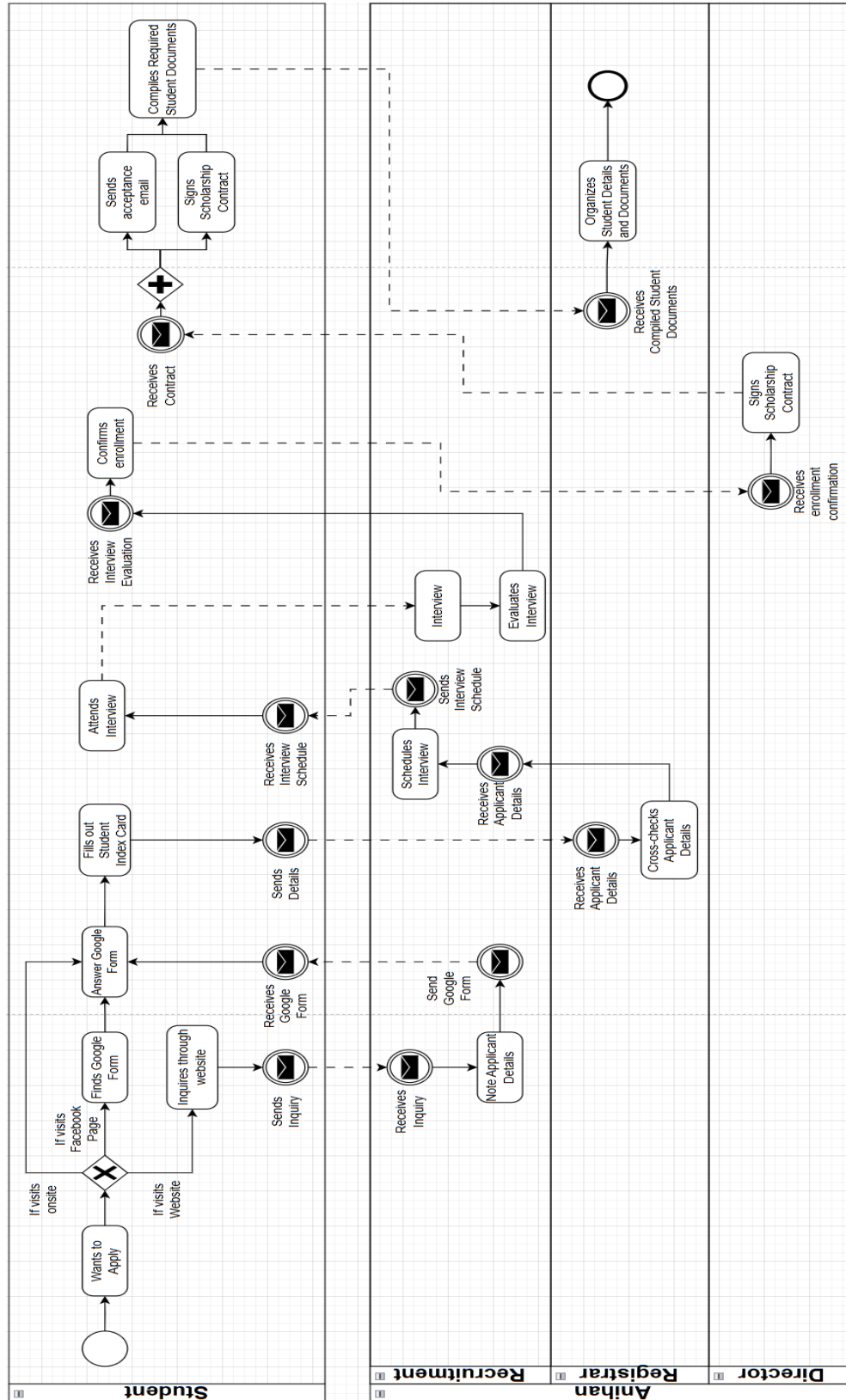


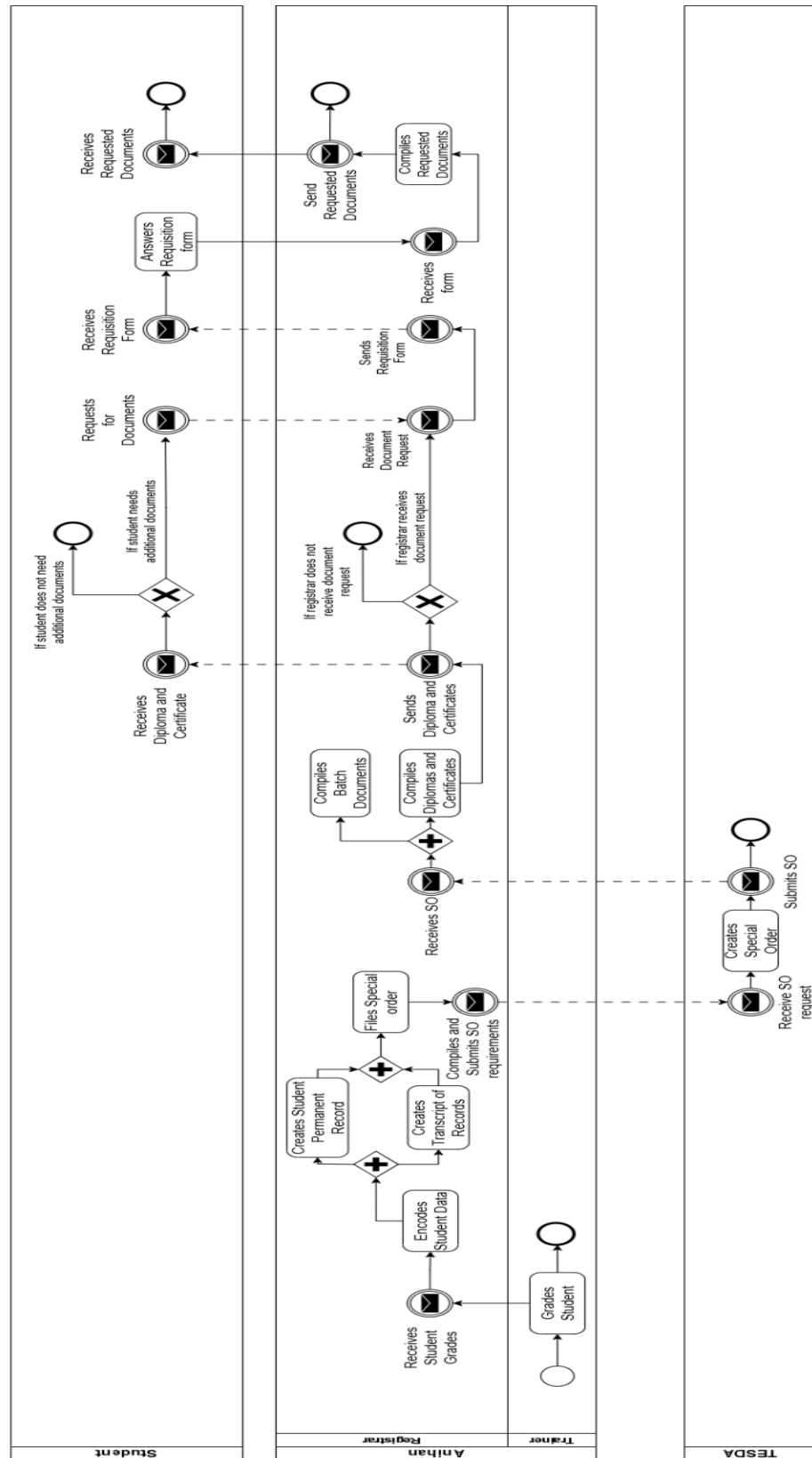
Figure 3.1.2.1 Enrollment BPMN

This BPMN shows the process of how a student enrolls in Anihan Technical School. The model consists of two pools: the student pool and the Anihan pool. The Anihan pool is divided into three lanes: recruitment, registrar, and director.

The process begins when a student wants to apply; they can encounter three ways to do so, the first being through the website. If the student inquires through the website, a message is sent to recruitment, who notes the details of the applicant and sends them a Google form to answer via email. The second way is if the student applies through their Facebook page, they would find the Google form link to answer. The last way is for the student to visit on-site, where they would be prompted to answer the Google form.

All of these ways lead to the student answering a Google form, and afterwards, they fill out an index card containing their details, requiring the other two methods for the student to visit on-site. The student's details are then sent to the registrar for cross-checking. The cross-checked details are received by recruitment, who then schedules and sends an interview to the student. After the student attends the interview, it is then evaluated by the recruitment department, and the evaluation is sent to the student. The student receives this evaluation and confirms their enrollment at Anihan Technical School.

The confirmation is sent to the director, who signs and sends the scholarship contract to the student. After which, the student sends their acceptance email, signed scholarship contract, and required documents to the registrar. The registrar then receives these documents and organizes the details and documents, ending their enrollment process.



The BPMN above shows the current student records management system that Aniham has. The model consists of three pools: the student pool, the Aniham pool, and the TESDA pool. The Aniham pool is divided into two lanes, being the registrar and the trainer.

The process begins with the trainer submitting the grades of the student to the registrar. The registrar encodes the student data, creating both a Student Permanent Record and a Transcript of Records. The registrar then files a special order request, submitting the compiled requirements to TESDA.

TESDA receives this request and creates a Special Order. After it is ready, they submit it back to the registrar. The registrar receives the Special Order and compiles the batch documents, diplomas, and certificates. The diplomas and certificates are sent to the student, after which they undergo the process of requesting their documents. The student requests their documents from the registrar, who sends them a requisition form to answer, and the registrar compiles the requested documents from the requisition form and sends them to the student.

3.2 System Design

This section discusses the system design of a proposed Student Records Management System (SRMS) for Aniham Technical School.

3.2.1 External Design

The Student record system will run on a dedicated server located within the registrar's office to ensure centralized data management. The server will be completely isolated from the public internet to protect student records from external cyber threats, operating strictly within a local network environment. The institution will need to provide a dedicated server unit equipped with a high-performance multi-core processor, Windows Server 2025 Standard edition, and a minimum of 16 gigabytes of RAM to handle concurrent database queries efficiently.

The server must be outfitted with a high-capacity Solid State Drive (SSD) for the operating system and database to ensure fast retrieval speeds, alongside a secondary Hard Disk Drive (HDD) serving as a redundant backup repository. The existing workstation in the registrar will connect to this server via a network switch using CAT6 ethernet cabling to ensure a stable, high-speed connection. Lastly, as a preventive measure against data corruption during power outages, the server setup requires the installation of a dedicated Uninterruptible Power Supply (UPS) unit.

3.2.2 Internal Design

The software architecture of the Student record system will utilize the Model-View-Controller (MVC) pattern. This design separates the application into three interconnected components, which ensures that the user interface, business logic, and data management are handled independently. This allows for easier maintenance and scalability of the system since they are separated. The user interface is the component that handles the presentation layer that the users interact with. This includes the web pages, forms, and dashboards created using HTML, CSS, and JavaScript. This component is responsible for displaying student data and capturing user inputs. The business logic acts as the intermediary between the user interface and the database. When the registrar submits a request, this component processes this command and applies the necessary validation rules. This then retrieves the specific data from the database and formats the result, and sends it back to the user interface for viewing. Finally, the database logic is the component that manages the data and business rules of the web application. It interacts directly with the database to execute queries—inserting new student profiles, updating grades, or retrieving archived documents—the database ensures that all data entering the system adheres to the defined structure before storage.

3.2.3 Data Structures of the Program

The data structures implemented in the SRMS serve as the foundation for the organization, storage, and processing of student-related information. The structures of this system contain object classes that represent the key entities. These are the students, users, TORs, and permanent records. The system also includes internal data structures, which are hash maps and lists, that facilitate data retrieval and operations of the system. Table 3.2.3.1 describes the data structures in detail.

Table 3.2.3.1 Data Structures of the Program

Data Structure	Type	Description	Purpose in the System
Student	Object Class	Stores the student's personal information, studentID, learnerID, student documents, date enrolled, academic year, course, batchID, subjects taken, grades, and attendance record	Manages student information for easy lookup and data update.
Batch	Object Class	This represents the specific group of students (e.g., "Batch 2024") containing batch_id, year, and program.	Groups students to facilitate bulk processing of Special Orders (SO) and minimizes repetitive data entry.
TranscriptOfRecords	Object Class	Links the student object data and encodes it in a TOR document template of the school	Receives student data and inputs them into the TOR template for printing or downloading, improving the encoding of student data
PermanentRecord	Object Class	Links the student object data and encodes it in a Permanent Record document template of the school	Receives student data and inputs them into the Permanent Record template for printing or downloading, improving the encoding of student data

User	Object Class	Stores account details: name, password hash, and assigned role	Enables authentication, access control, and role-based permissions
TransactionLog	Object Class	Records user actions (ex., Update grades, input new students)	Provides accountability, traceability, and security within the system.
Document	Object Class	Links a student to their digitized records, containing document_type and the secure file_path on the server.	Handles the retrieval of scanned files (e.g., PSA, Form 137) without overloading the database storage.
HashMap<String, Students>	Hash Table	Key-value structure using studentID as key and Student object as value.	Provides constant time data retrieval of Students to improve search efficiency
HashMap<String, User>	Hash Table	Key-value structure mapping usernames to User objects.	Provides security and role authentication of users upon login
ArrayList<Student>	List	Stores the complete list of students	Facilitates the attendance, enrollment, and program completion reports of students, and can be printed or downloaded. Students can be filtered based on the studentID, learnerID, and batchID
ArrayList<TranscriptOfRecords>	List	Stores the complete list of TOR documents	Facilitates the verification that all students have a TOR
ArrayList<PermanentRecord>	List	Stores the complete list of PermanentRecord documents	Facilitates the verification that all students have a Permanent Record
ArrayList<TransactionLog>	List	Stores the complete list of TransactionLogs in chronological order	Helps with the review of activities in the system
ArrayList<ErrorLog>	List	Stores chronological error logs	Helps with the debugging of the system

Table 3.2.3.1 shows the system integration of both object-oriented and non-object-oriented data structures to achieve accuracy, efficiency, and scalability. Student, User, Batch, Document, TranscriptOfRecords, and PermanentRecord object classes are the core system entities. These object classes encapsulate different key attributes, variables, and behaviors to ensure the data is organized and manageable. Hash maps and array lists are also implemented as internal data structures to handle recurring data retrieval and storage of information, respectively.

3.2.4 Use Case Diagrams

The use cases of the system are discussed here through the use of use case diagrams. Figure 3.2.4.1 illustrates the interactions that the system admin will have with the system.

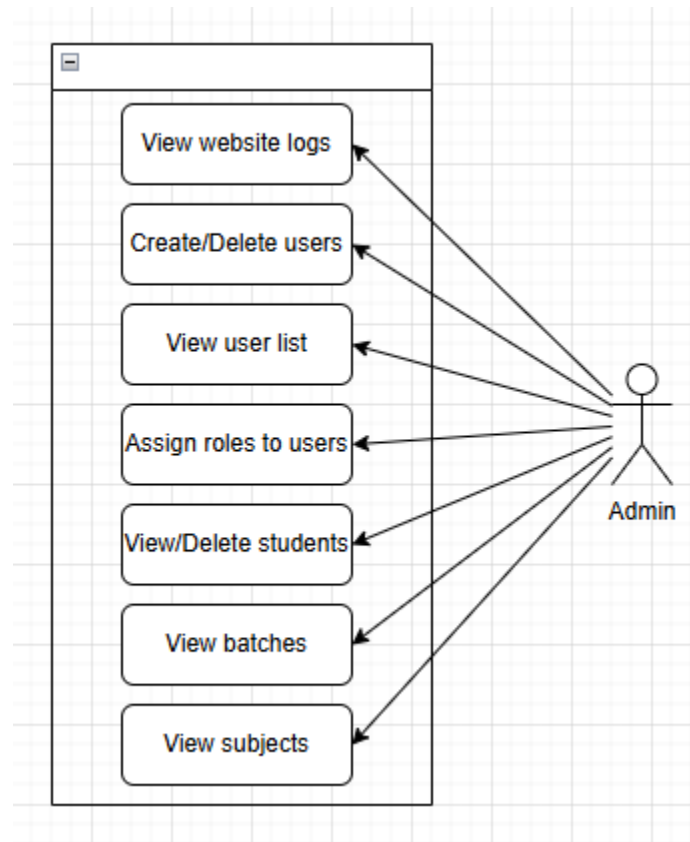


Figure 3.2.4.1 Admin Use Case

Figure 3.2.4.1 shows that the system includes viewing the website logs, creating and deleting users of the system, viewing the user list and assigning roles to them, viewing and deleting students, as well as viewing batches and subjects. The role of the admin is to oversee the operation of the system and assign the roles and permissions of the different users, including the registrar, trainer, and student.

Figure 3.2.4.2 illustrates the interactions the Registrar will have with the system.

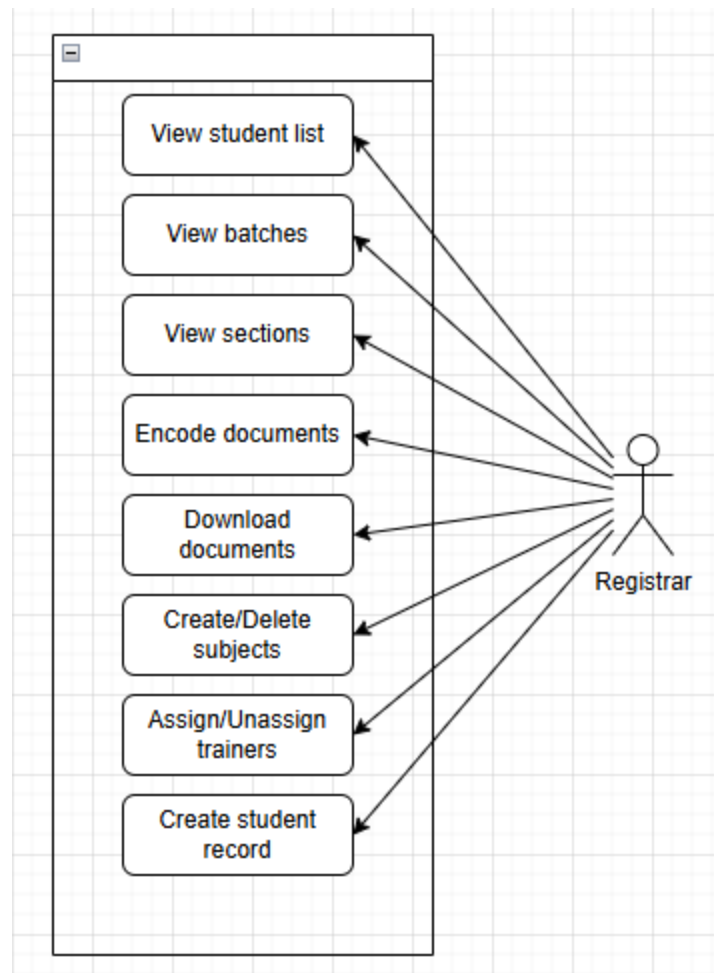


Figure 3.2.4.2 Registrar Use Case

In Figure 3.2.4.2, the registrar has the functions of viewing the student list, batches, and sections, encoding and downloading of documents, creating and deleting subjects, assigning trainers, and creating student records. This is where the main functionalities of the system will lie.

Figure 3.2.4.3 shows the use case of the trainer interacting with the system.

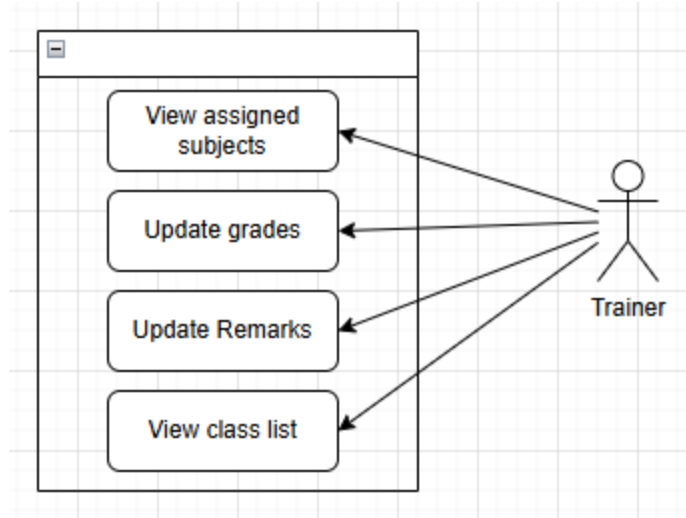


Figure 3.2.4.3 Trainer User Case

Figure 3.2.4.3 illustrates the interaction the trainer will have with the system, including the viewing of assigned subjects and class list, and updating of grades and remarks.

Figure 3.2.4.4 shows the use case of students interacting with the system.

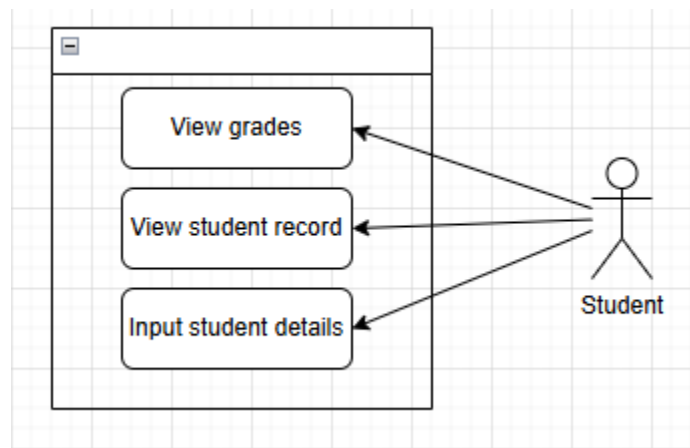


Figure 3.2.4.4 Student Use Case

Figure 3.2.4.4 illustrates the interaction the students will have with the system, including the viewing of their grades and student records, and inputting of their student details.

3.2.5 Class Diagram

Different classes will be created for the system in the object-oriented approach. Figure 3.2.5.1 is a class diagram for the system.

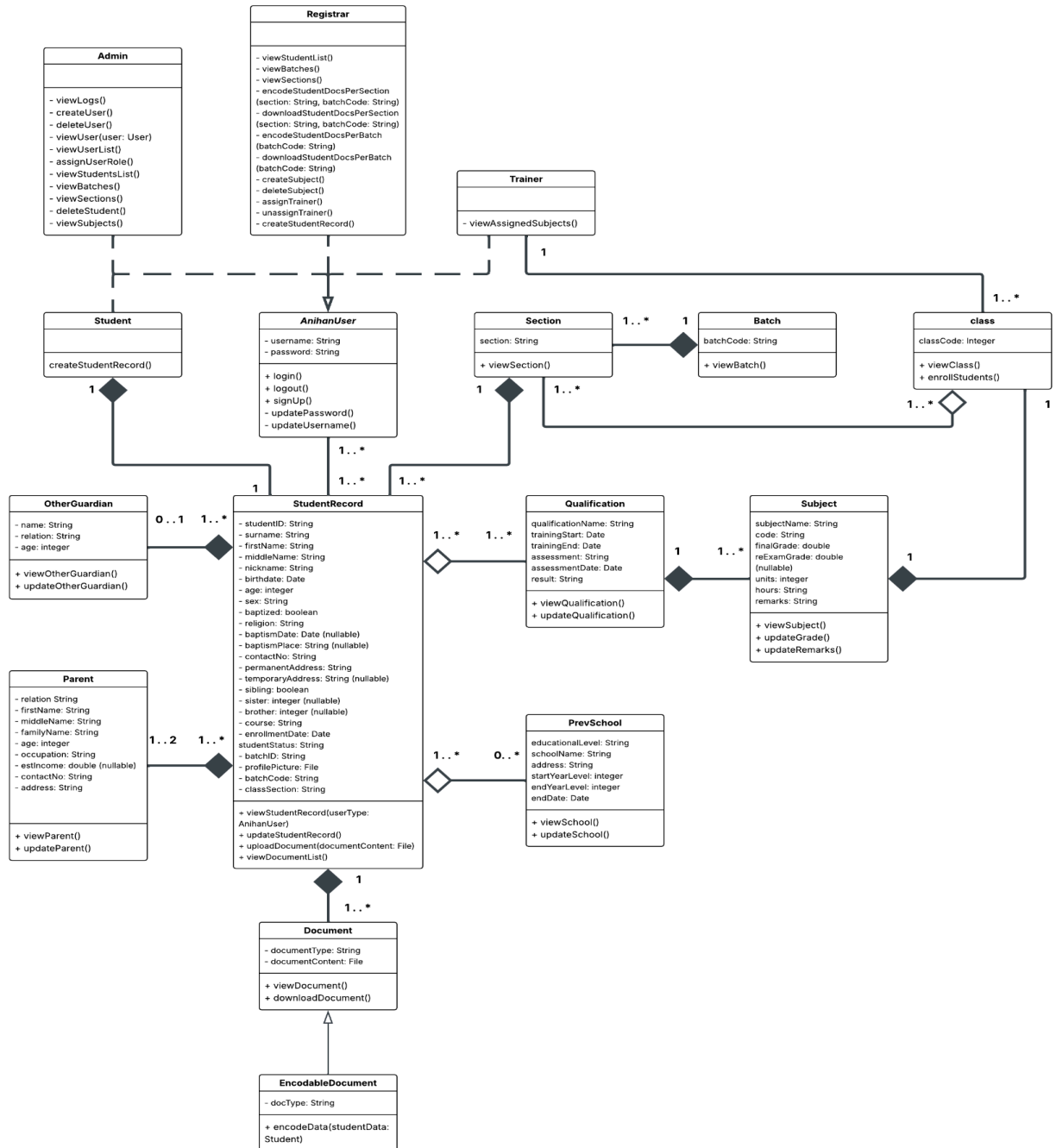


Figure 3.2.5.1 Student Records Management System Class Diagram

In Figure 3.2.5.1, the central class is StudentRecord, which functions as the container for all data referring to the students. The attributes and classes that compose this class are based on the information required to be inputted into the student index card of the school. The StudentRecord encapsulates classes Parent, OtherGuardian, which pertain to the family members or related guardians of the student. A StudentRecord may have none to two parents, indicating that they have no parents, a mother, a father, or both. OtherGuardian is optional, but the StudentRecord can only have a relative or legal guardian. These three classes have a one-to-many relationship to the StudentRecord, as there could be more than one StudentRecord associated with them, which also makes these classes dependent on the existence of at least one StudentRecord.

StudentRecord encapsulates Document, which is the digitized student document requirements that are submitted for enrollment. EncodableDocument is a subclass of Document, which pertains to the TOR and Permanent Record of the student that is to be encoded by taking information from the StudentRecord. The Document has a many-to-one dependent relationship with StudentRecord, as these documents only refer to a specific student in the system.

StudentRecord encapsulates the PrevSchool class, which contains information regarding the student's previous schools attended from primary education to college. This class has none or many-to-many relationship with StudentRecord, as the student may not have attended any previous school or may have skipped an educational level. The class can also exist independently of the StudentRecord, since the school would need to keep records of the schools that students have attended.

The Qualification class contains the information pertaining to the qualifications, specific to the course taken, of the student. It encapsulates the qualification assessment and training start and end dates. It has an independent many-to-many relationship with StudentRecord since a qualification could exist independently from the student record. It also encapsulates the Subject class, which contains data on specific subjects to complete the qualifications. It has a dependent many-to-one relationship with Qualification, as multiple subjects exist for a specific qualification.

The SRMS will have four user roles. These are the Student, Admin, Registrar, and Trainer classes. These classes implement the AnihanUser interface, which provides the signup, login, and signout capabilities. These users could also update the username, password, and email of their account. The role of the Student class is to allow students to create their StudentRecord object and view it. The Admin is the one responsible for managing the system. Apart from seeing the classes, sections, batches, students, users, and subjects stored in the system, they are to create the user accounts for Registrars, Trainers, and other Admins. They could also delete users in the system. The Registrar is responsible for encoding and downloading the documents of the student. The Trainer is responsible for managing a class, which is assigned to them by the Registrar.

3.2.6 Sequence Diagram

The sequence diagram illustrates the dynamic interactions between the system's internal objects, detailing the time-ordered sequence of events and messages required to execute specific functions.

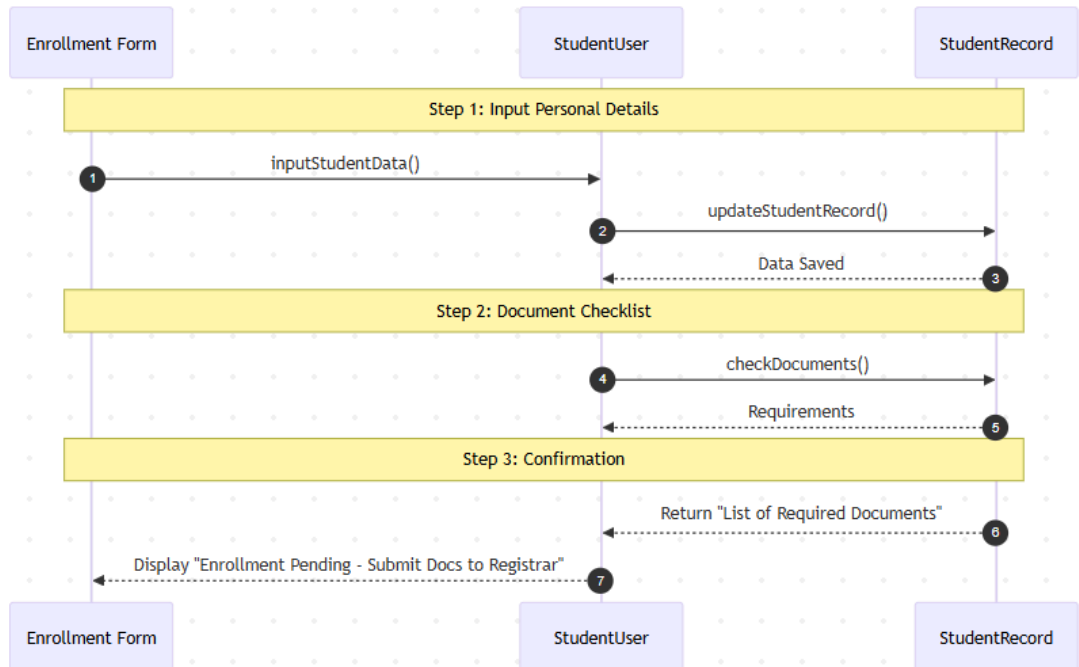


Figure 3.2.6.1 Enrollment Sequence Diagram

The sequence diagram in Figure 3.2.6.1 illustrates the sequential logic of the student enrollment process. The workflow initiates when the `inputStudentData()` method is triggered via the `StudentUser` interface, which subsequently instantiates a new profile using the `addNewStudentRecord()` method in the `StudentRecord` class. The system then saves the personal data of the student and executes a verification step via `checkDocument()` to identify the necessary requirements for the enrollee. The system then sends the list of required documents to be submitted, prompting the interface to display a status message indicating that the enrollment is pending the physical submission of documents to the registrar.

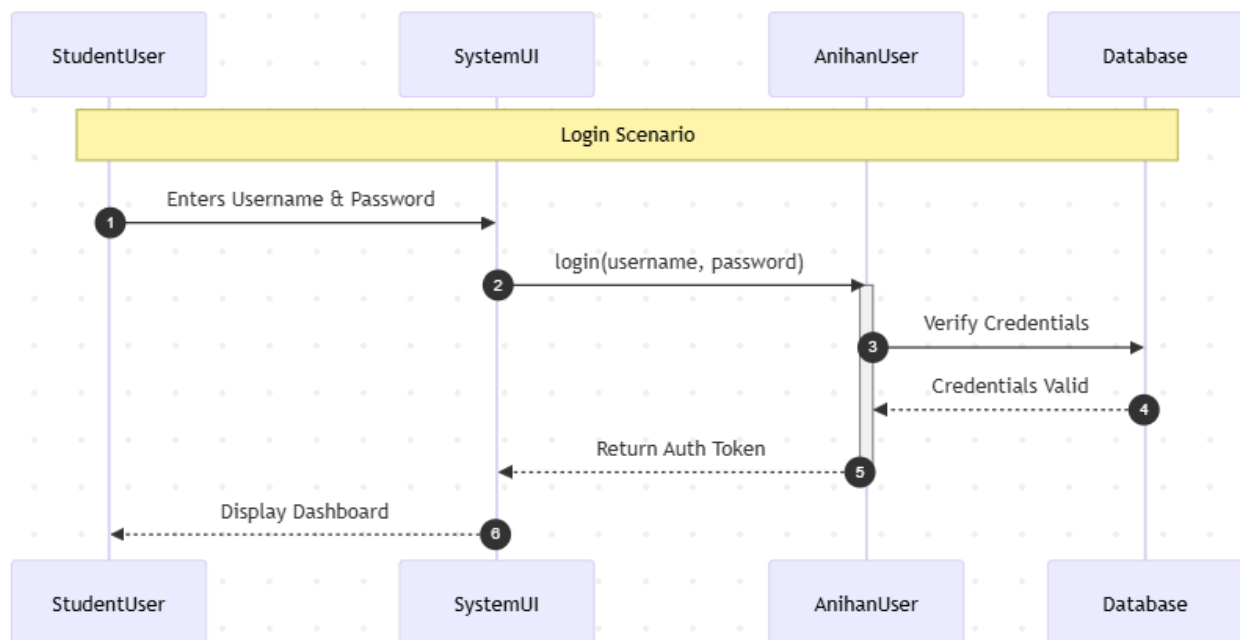


Figure 3.2.6.2 User Log-In Authentication Sequence Diagram

Figure 3.2.6.2 illustrates the authentication process of users' accounts. The `StudentUser` acts as the primary actor initiating the request through a `SystemUI`. The UI calls the `login()` operation defined in the `AnihanUser` class, which checks against a `DataBase` to verify the username and password. Once validated, the system grants access to the user.

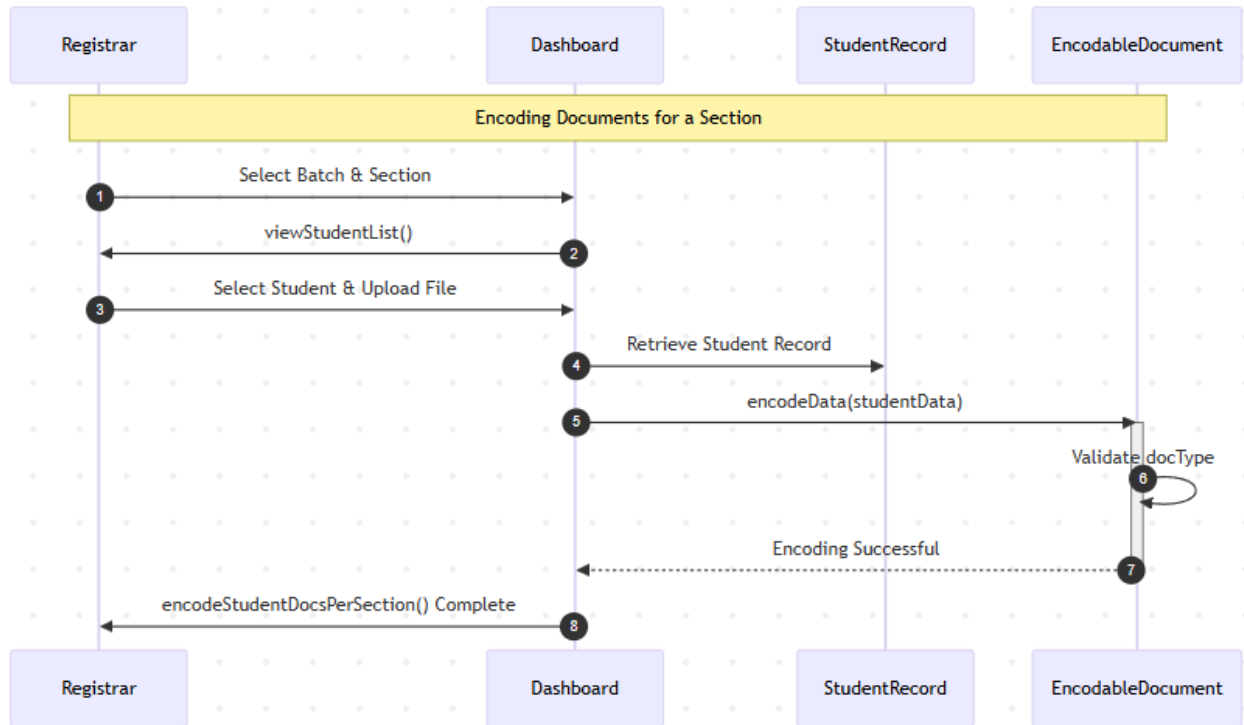


Figure 3.2.6.3 Encoding of Student Documents Sequence Diagram

The sequence diagram seen in Figure 3.2.6.3 shows the specific responsibility of the Registrar, who is tasked with batch processing documents. The flow begins with the Registrar viewing the student list, then invoking the `encodeStudentDocsPerSection` logic. The diagram shows the interaction between the Registrar and the `EncodableDocument` class, where the `encodeData` method is triggered to attach the physical file to the student's digital record.

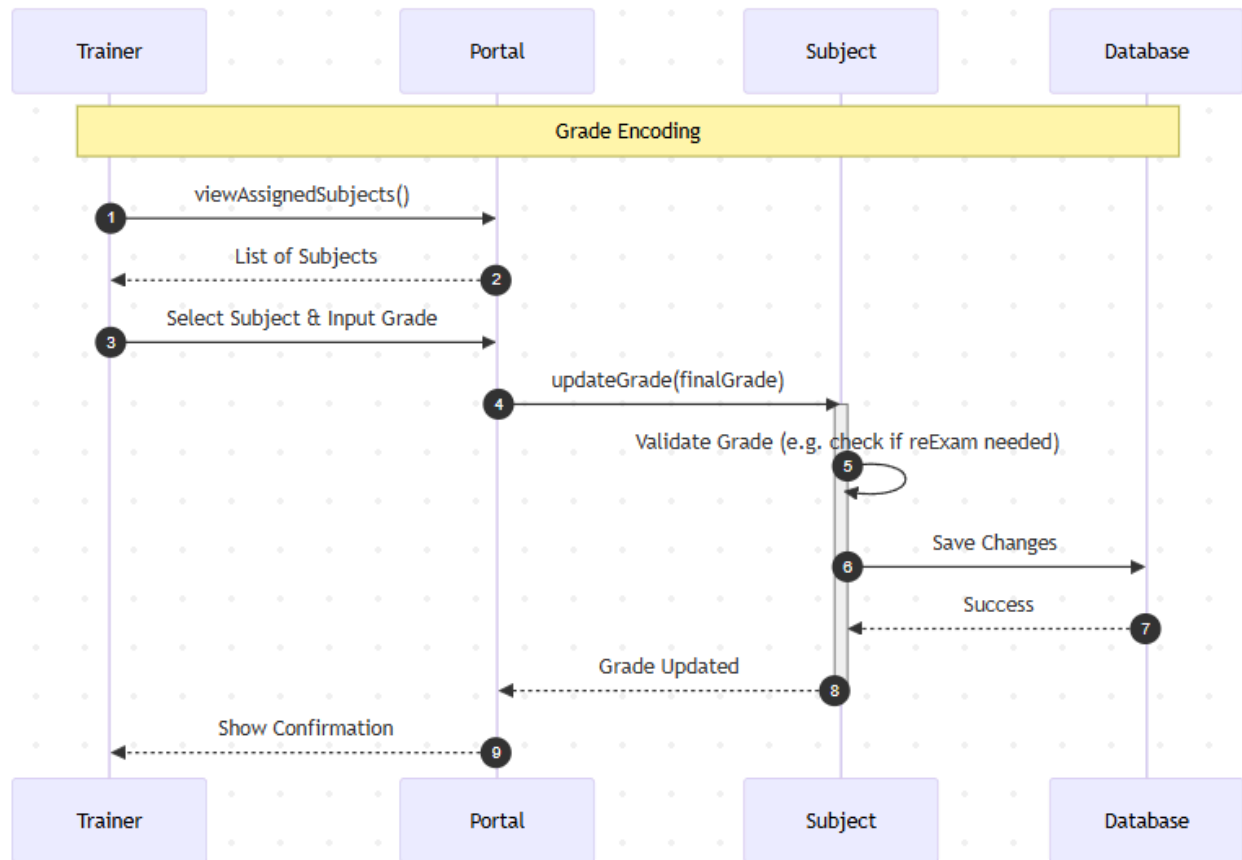


Figure 3.2.6.4 Updating Student Grades Sequence Diagram

The sequence diagram shown in Figure 3.2.6.4 depicts the academic workflow for a Trainer. It starts with the Trainer calling `viewAssignedSubjects()` to see their load. When they submit a grade, the system triggers the `updateGrade()` method within the Subject class. The diagram logic also hints at the internal check for the `finalGrade` or `reExamGrade` attributes before committing the data to the database.

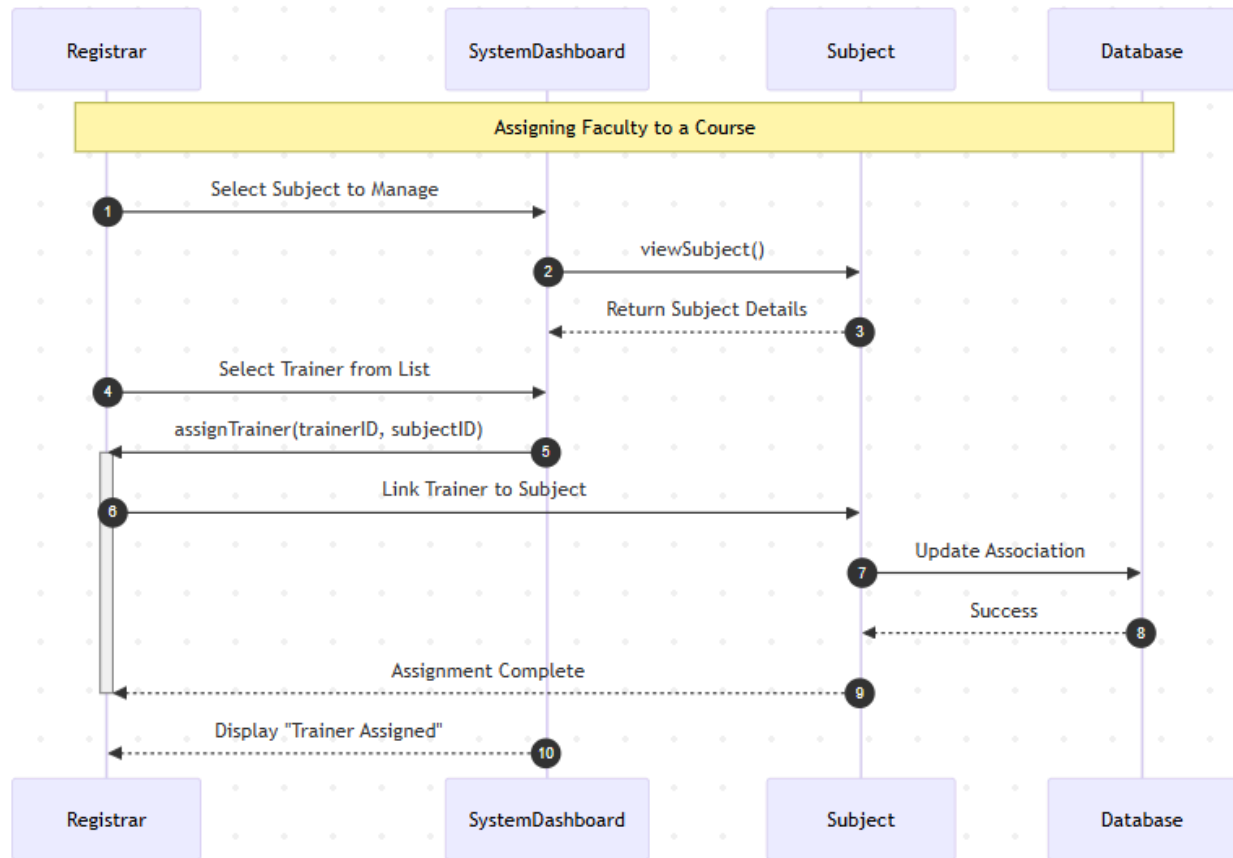


Figure 3.2.6.5 Registrar Assigning a Trainer to a Subject Sequence Diagram

The sequence diagram seen in Figure 3.2.6.5 depicts the administrative workflow where the Registrar manages faculty assignments. The process begins with the Registrar selecting a specific subject through the SystemDashboard, which retrieves the subject details via `viewSubject()`. The Trainer then is selected from the list and the system invokes the `assignTrainer()` method, which triggers the Subject object to update its internal state by linking the specified trainer ID. This new association is then committed to the Database, and upon a successful transaction, the system provides confirmation back to the Registrar, ensuring the academic load is correctly distributed.

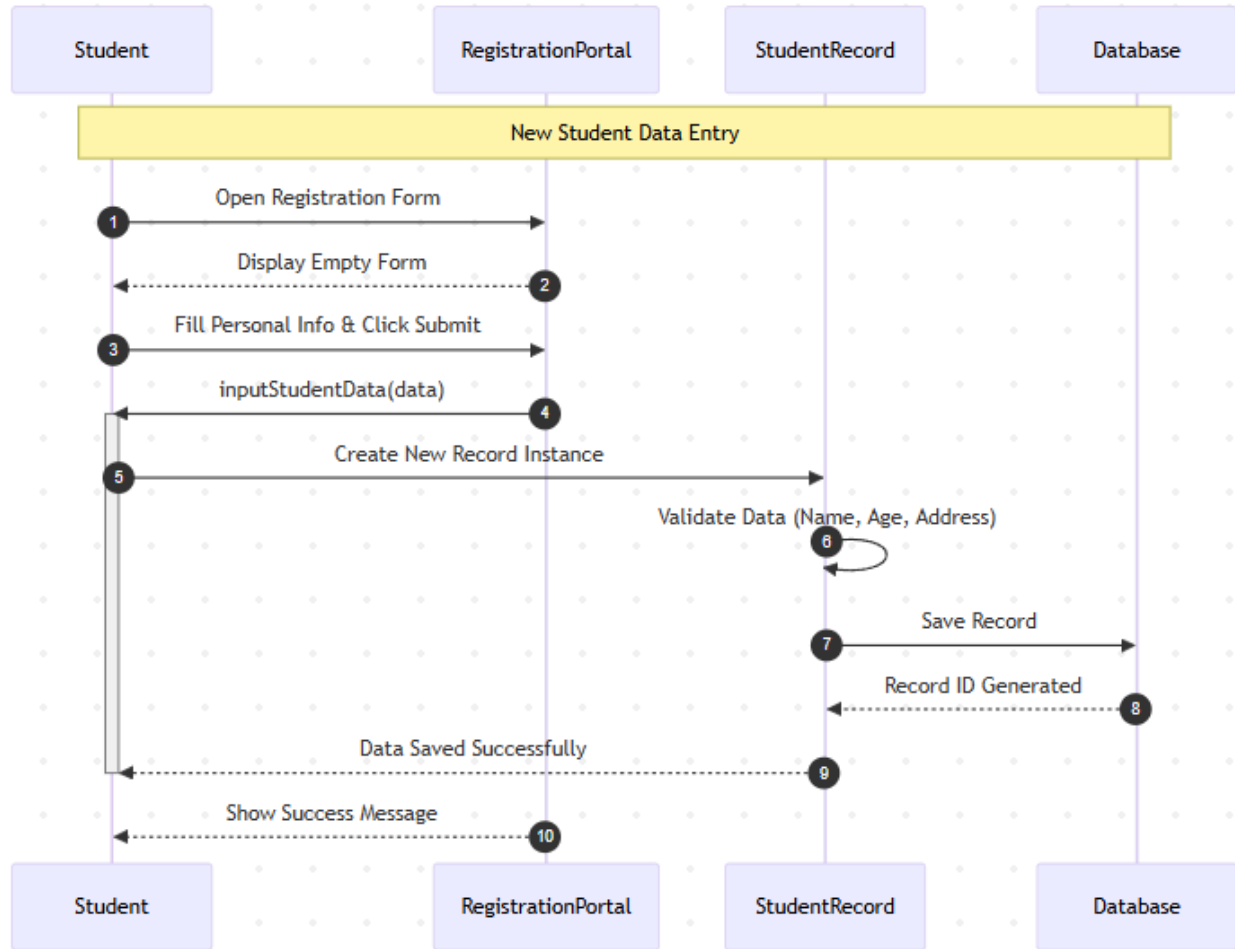


Figure 3.2.6.6 Student Inputting Personal Data Sequence Diagram

Figure 3.2.6.6 illustrates the Student inputting their own personal information into the system. The interaction starts when the Student accesses the RegistrationPortal and submits their personal information, triggering the `inputStudentData()` method. The system then creates a temporary instance of a `StudentRecord`, which internally validates critical fields before attempting to store the data. Once the Database confirms the record is saved and a unique ID is generated, the system completes the loop by displaying a success message to the Student.

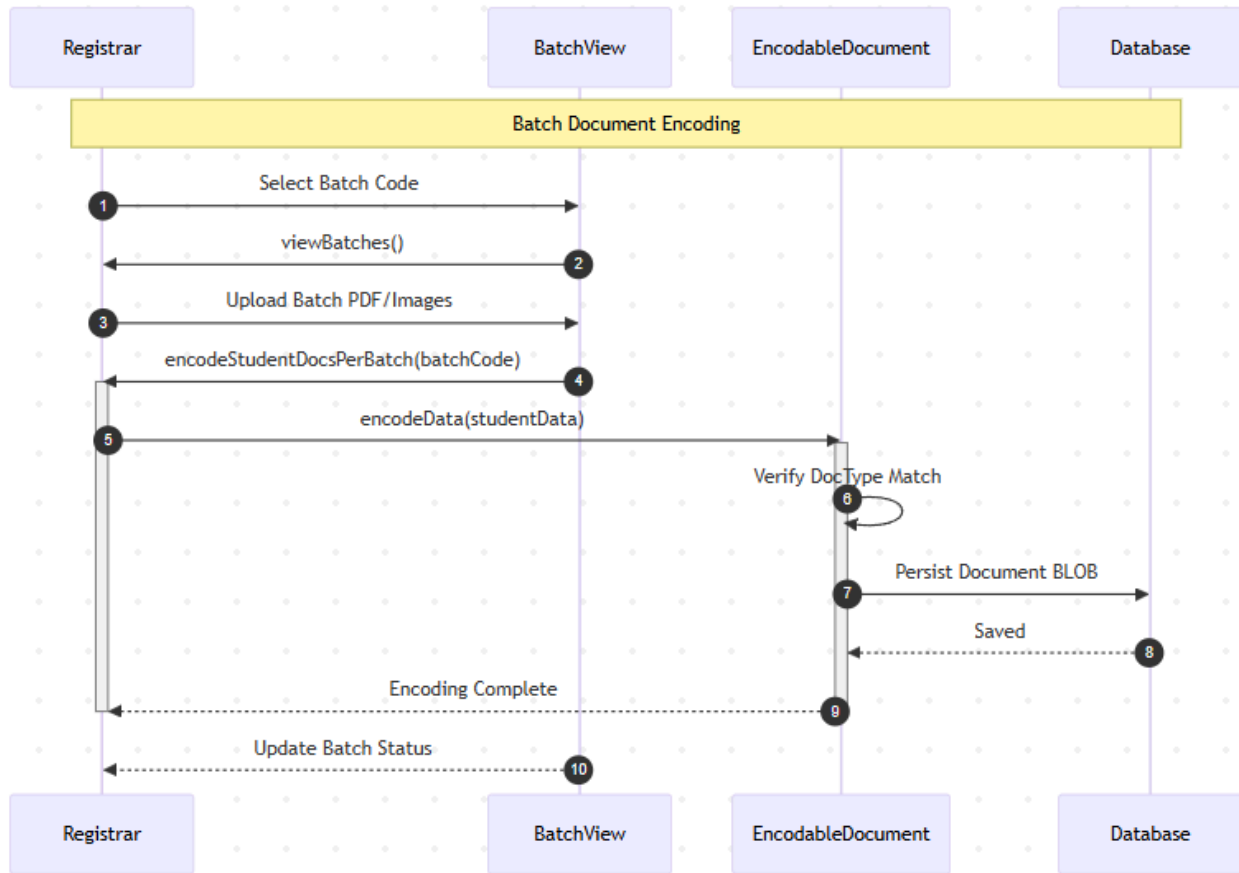


Figure 3.2.6.7 Registrar Encoding Batch Documents Sequence Diagram

The sequence diagram shown in Figure 3.2.6.7 models the bulk processing efficiency required for the Registrar when handling student documentation. The flow initiates when the Registrar selects a specific batch code via the BatchView interface and uploads the corresponding digital files—PDFs or images. The system executes the `encodeStudentDocsPerBatch()` method, which delegates the actual processing to the EncodableDocument class; this class verifies the document type and encodes the data before storing the binary large object (BLOB) into the Database. The sequence is done when the interface updates the batch status, confirming that all physical records for that batch have been successfully digitized.

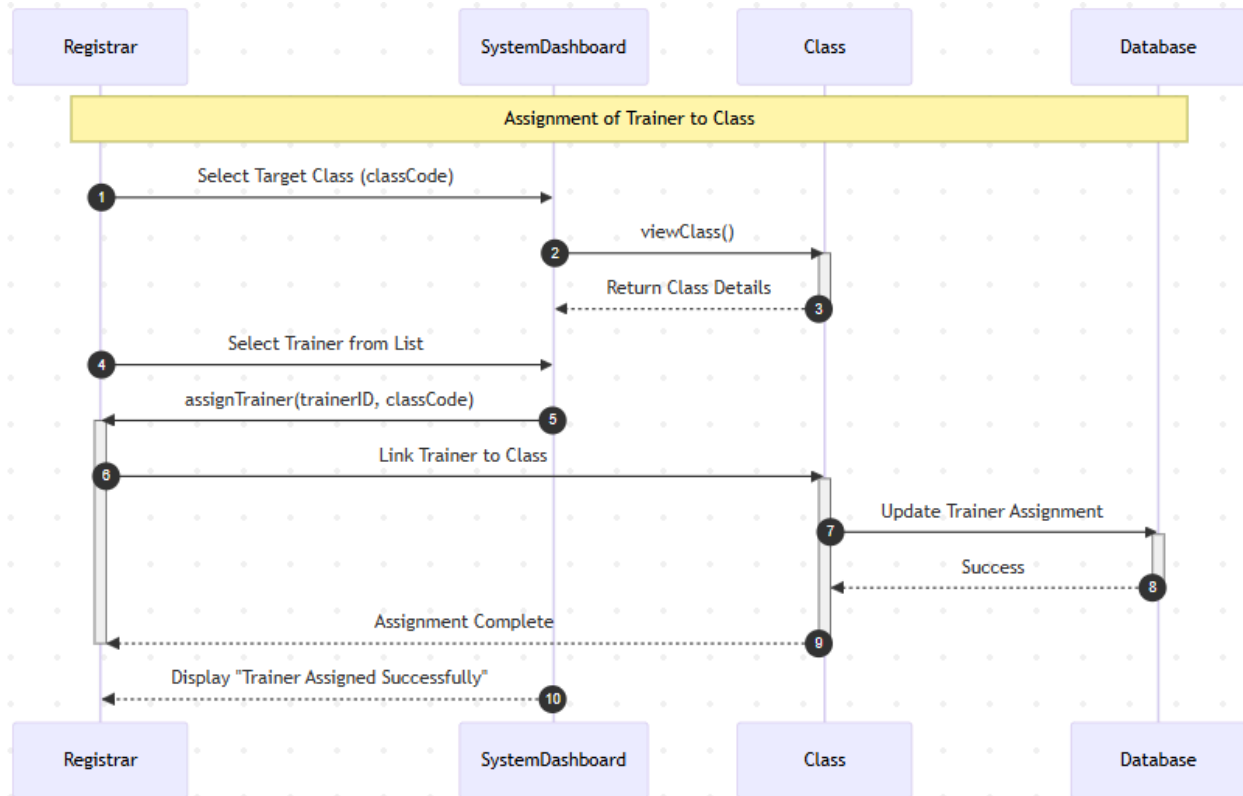


Figure 3.2.6.8 Register Assigning Trainer to a Class Sequence Diagram

The sequence diagram seen on Figure 3.2.6.8 illustrates the process of resource allocation where the Registrar assigns a faculty member to a specific class section. The interaction begins with the Registrar selecting a class via the dashboard, triggering the `viewClass()` method to retrieve the Class object's details. Upon selecting a Trainer, the `assignTrainer()` method is invoked, which establishes a logical link between the Trainer and the Class in the system. This assignment is then committed to the Database ensuring that when the Trainer later logs in and calls their new `viewAssignedClasses()` method, this specific class will successfully appear in their schedule.

3.2.7 Entity Relationship Diagram

Figure 3.2.7.1 is an Entity Relationship Diagram (ERD) that shows the entities present and their relationship with one another in the database of the SRMS.

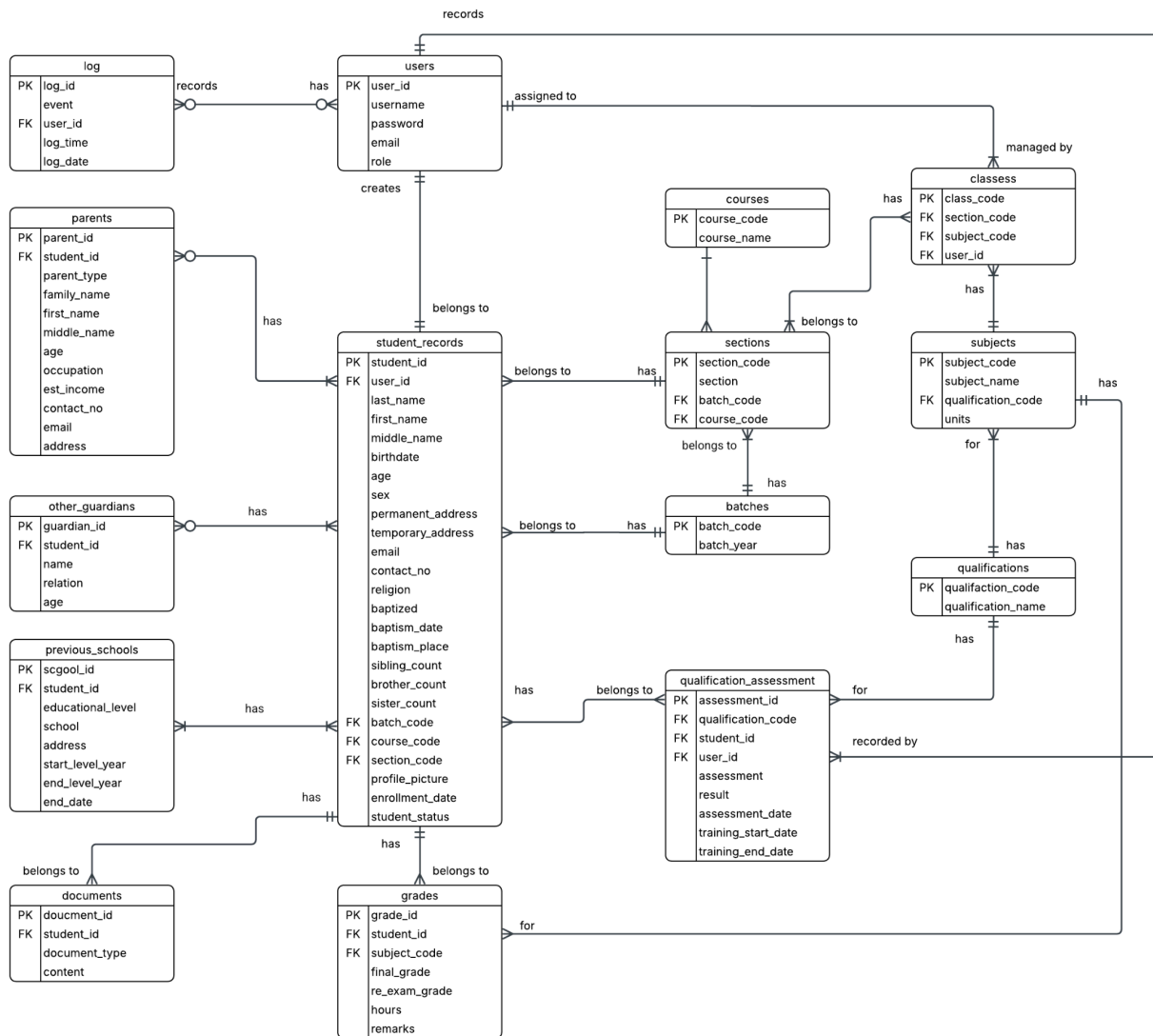


Figure 3.2.7.1 Entity Relationship Diagram of the SRMS

Based on the given role to a user, it determines the specific relationship the entity has with the others. Student can create a student record, trainers are assigned to manages one or many classes while also providing the grades for students, and registrar provides the qualification assessments for all students.

Each student record can have none or more parents, none or one guardian, none or more previous schools that they have attended, and has many documents. Each student belongs to a section, that is also part of the batch for that academic year. Each section attends a specific class in order to get the qualification for a specific skill as determined by their course.

Every interaction and event in the SRMS is recorded as transaction logs. The logs contain a description of the event, the person responsible for the event, and the time and date of the recording. This ensures proper documentation and error handling, should it ever occur, within the system, to be monitored by the admin.

3.2.8 Data Dictionary

Tables 3.2.8.1 to 3.2.8.15 are the data dictionaries for each of the entities as illustrated by the ERD (see Figure 3.2.7.1).

Table 3.2.8.1 Data Dictionary for student_records

Field Name	Description	Data Type	Constraints	Data Default
student_id	Unique identifier for student	VARCHAR	Primary key	
user_id	Identifier of the associated student user	INT	Foreign key → users	
last_name	Last name of the student	VARCHAR	Not null	
first_name	First name of the student	VARCHAR	Not null	
middle_name	Middle name of the student	VARCHAR	Not null	
birthdate	Birthdate of the student	DATE	Not null	
age	Age of the student	INT	Not null	
sex	Sex of the student	VARCHAR	Not null	

permanenet_address	Permanent address of the student	VARCHAR	Not null	
temporary_address	Temporary address of the student	VARCHAR	Nullable	
email	Email of the student	VARCHAR	Not null	
contact_no	Contact number of the student	VARCHAR	Not null	
religion	Religion of the student	VARCHAR	Not null	
baptized	Indicates if the student has been baptized	BOOLEAN	Not null	
baptism_date	Baptism date of the student	DATE	Nullable	
baptism_place	Baptism place of the student	VARCHAR	Nullable	
sibling_count	Sibling count of the student	INT	Not null	
brother_count	Brother count of the student	INT	Nullable	
sister_count	Sister count of the student	INT	Nullable	
batch_code	Identifier for the batch of the student	VARCHAR	Foreign key → batches	
course_code	Identifier for the course of the student	VARCHAR	Foreign key → courses	
section_code	Identifier for the section of the student	INT	Foreign key → sections	
profile_picture	Profile picture of the student	BLOB	Not null	Default Image
enrollment_date	Enrollment date of the student	DATE	Nullable	

student_status	Student status within the school of the student	VARCHAR	Not null	Enrolling
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Table 3.2.8.2 Data Dictionary for parents

Field Name	Description	Data Type	Constraints	Data Default
parent_id	Unique identifier for the parent	INT	Primary key, auto-increment	
student_id	Identifier of the associated student of the parent	VARCHAR	Foreign key → student_records	
relation	Indicates the relation of the parent to the student	VARCHAR	Not nullable	
family_name	Family name of the parent	VARCHAR	Not nullable	
first_name	First name of the parent	VARCHAR	Not nullable	
middle_name	Middle name of the parent	VARCHAR	Not nullable	
age	Age of the parent	INT	Not nullable	
occupation	Occupation of the parent	VARCHAR	Not nullable	
est_income	Estimated income of the parent	DOUBLE	Nullable	
contact_no	Contact number of the parent	VARCHAR	Not nullable	
email	Email of the parent	VARCHAR	Not nullable	
address	Address of the parent	VARCHAR	Not nullable	

Table 3.2.8.3 Data Dictionary for other_guardians

Field Name	Description	Data Type	Constraints	Data Default
guardian_id	Unique identifier	INT	Primary key,	

	for the guardian		auto-increment	
student_id	Identifier of the associated student of the guardian	VARCHAR	Foreign key → student_records	
relation	Relation of the guardian to the student	VARCHAR	Not nullable	
name	Name of the guardian	VARCHAR	Not nullable	
age	Age of the guardian	INT	Not nullable	

Table 3.2.8.4 Data Dictionary for documents

Field Name	Description	Data Type	Constraints	Data Default
document_id	Unique identifier of the document	INT	Primary key, auto-increment	
student_id	Identifier of the associated student of the document	VARCHAR	Foreign key → student_records	
document_type	Document type of the document	VARCHAR	Not nullable	
content	Content of the document	LOB	Not nullable	

Table 3.2.8.5 Data Dictionary for grades

Field Name	Description	Data Type	Constraints	Data Default
grade_id	Unique identifier for the grade	INT	Primary key, auto-increment	
student_id	Identifier of the associated student of the grade	VARCHAR	Foreign key → student_records	
subject_code	Identifier of the associated subject of the grade	VARCHAR	Foreign key → subjects	
final_grade	Final grade of the student	FLOAT	Not nullable	0

re_exam_grade	Re-exam grade of the student	FLOAT	Nullable	
hours	The number of hours the student took to get this grade	INT	Not nullable	
remarks	Remarks of the trainer for the grade	VARCHAR	Not nullable	

Table 3.2.8.6 Data Dictionary for sections

Field Name	Description	Data Type	Constraints	Data Default
section_code	Unique identifier for the section	INT	Primary key, auto-increment	
section	Name of the section	VARCHAR	Not nullable	
batch_code	Identifier of the associated batch of the section	VARCHAR	Foreign key → batches	
course_code	Identifier of the associated course of the section	VARCHAR	Foreign key → courses	

Table 3.2.8.7 Data Dictionary for batches

Field Name	Description	Data Type	Constraints	Data Default
batch_code	Unique identifier of the batch	VARCHAR	Primary key	
batch_year	Academic year of the batch	DATE	Not nullable, format (YYYY)	

Table 3.2.8.8 Data Dictionary for courses

Field Name	Description	Data Type	Constraints	Data Default
course_code	Unique identifier of the course	VARCHAR	Primary key	
course_name	Name of the course	VARCHAR	Not nullable	

Table 3.2.8.9 Data Dictionary for qualifications

Field Name	Description	Data Type	Constraints	Data Default
qualification_code	Unique identifier of the qualification	INT	Primary key, auto-increment	
qualification_name	Name of the qualification	VARCHAR	Not nullable	

Table 3.2.8.10 Data Dictionary for qualification_assessment

Field Name	Description	Data Type	Constraints	Data Default
assessment_id	Unique identifier for the assessment	INT	Primary key, auto-increment	
qualification_code	Identifier of the associated qualification of the assessment	INT	Foreign key → qualifications	
student_id	Identifier of the associated student of the assessment	VARCHAR	Foreign key → student_records	
user_id	Identifier of the associated assessor of the assessment	INT	Foreign key → users	
assessment	Remarks of the assessment	VARCHAR	Not nullable	
result	Result of the assessment	VARCHAR	Not nullable	
assessment_date	Date of the assessment	DATE	Not nullable	
training_start_date	Start date of the training for the qualification	DATE	Not nullable	
training_end_date	End date of the training for the qualification	DATE	Not nullable	

Table 3.2.8.11 Data Dictionary for subjects

Field Name	Description	Data Type	Constraints	Data Default
subject_code	Unique identifier	VARCHAR	Primary key	

	of the subject			
subject_name	Name of the subject	VARCHAR	Not nullable	
qualification_code	Identifier of the associated qualification	INT	Foreign key → qualifications	
units	Number of units the subject has	INT	Not nullable	

Table 3.2.8.12 Data Dictionary for classes

Field Name	Description	Data Type	Constraints	Data Default
class_code	Unique identifier of the class	INT	Primary key, auto-increment	
section_code	Identifier of the section assigned to this class	INT	Foreign key → sections	
subject_code	Identifier of the subject of this class	VARCHAR	Foreign key → subjects	
user_id	Identifier of the assigned trainer user	INT	Foreign key → users	

Table 3.2.8.13 Data Dictionary for users

Field Name	Description	Data Type	Constraints	Data Default
user_id	Unique identifier of the user	INT	Primary key, auto-increment	
username	Username of the user	VARCHAR	Not nullable	
password	Password of the user	VARCHAR	Not nullable	
email	Email of the user	VARCHAR	Not nullable	
role	Role of the user in the system	VARCHAR	Not nullable	

Table 3.2.8.14 Data Dictionary for log

Field Name	Description	Data Type	Constraints	Data Default
log_id	Unique identifier of the log	INT	Primary key, auto-increment	
event	Description of the event that was recorded	VARCHAR	Not nullable	
user_id	Identifier of the associated user who conducted the event	INT	Foreign key → users	
log_time	The time the log was recorded	TIME	Not nullable	
log_date	Date the log was recorded	DATE	Not nullable	

Table 3.2.8.15 Data Dictionary for previous_school

Field Name	Description	Data Type	Constraints	Data Default
school_id	Unique identifier for the school	INT	Primary key, auto-increment	
student_id	Identifier of the student who attended the school	VARCHAR	Foreign key → student_records	
educational_level	The educational level the student took while enrolled in this school	VARCHAR	Not nullable	
school	The name of the school	VARCHAR	Not nullable	
address	Address of the school	VARCHAR	Not nullable	
start_level_year	The year the student's stay in the school for this educational level started	DATE	Not nullable, format (YYYY)	
end_level_year	The year the	DATE	Not nullable,	

	student's stay in the school for this educational level ended		format (YYYY)	
end_date	The date the student finished studying in this school	DATE	Not nullable	

3.2.9 Functional Requirements

The functional requirements detail the specific capabilities and behavioral specifications of the Student Record Management System, categorized by user roles to address the operational needs of the school.

Admin

1. The Admin is the primary administrator of the system. In order to secure the system and manage faculty access, the Administrator must validate and approve the accounts created by Trainers before they can access the system.
 - 1.1 The system shall allow the Admin to assign specific roles (Admin, Registrar, Trainer) to users—assignUserRole.
 - 1.2 The system shall allow the Admin to create and delete users—createUser, deleteUser.
 - 1.3 The system shall allow the Admin to view logs of system activities—viewLogs.
 - 1.4 The system shall allow the Admin to delete student records from the database—deleteStudent.

Registrar

1. The Registrar is responsible for maintaining the integrity of student data. This includes the ability to modify student information to correct errors and the digitization of physical documents to create a centralized record.

- 1.1 The system allows the Registrar to search for any student record using the Student ID or name.
- 1.2 The system allows the Registrar to edit the personal information and contact details of an existing student.
- 1.3 The system allows the Registrar to upload scanned documents—PSA, Form 137—to a student’s profile.
- 1.4 The system allows the Registrar to view the completion status of a student’s document checklist.
- 1.5 The system shall allow the Registrar to create and delete subjects—createSubject, deleteSubject.
- 1.6 The system shall allow the Registrar to download and encode student documents by batch or by section.
- 1.7 The system shall allow the Registrar to assign or unassign a Trainer to a specific subject.
2. To ensure the active database remains efficient and uncluttered, the Registrar needs to segregate the records of alumni from the current student body.
 - 2.1 The system allows the Registrar to change a student’s status between “Enrolling,” “Active,” and “Graduated”.
 - 2.2 The system automatically sorts and filters the records marked as “Graduated” and “Active”.

Student

1. During the enrollment period, the Student is responsible for the initial data entry. In order to reduce the manual workload of the Registrar, the Student must create an account and populate their own personal information into the system.
 - 1.1 The system allows the Student to register a new account during the open enrollment window.
 - 1.2 The system shall allow a Student to input their own student data—inputStudentData—upon initial registration or profile update.

- 1.3 The system requires the Student to fill out all mandatory data fields before submission.
- 1.4 The system displays a confirmation status of "Pending Approval" once the data is submitted.
- 2. Once enrolled, the Student requires read-only access to their records to verify their information and monitor their document submission status without the ability to alter permanent academic data.
 - 2.1 The system allows the Student to log in using their verified Student ID.
 - 2.2 The system allows the Student to view their personal profile and encoded demographic data.
 - 2.3 The system prohibits the Student from editing their name or personal details after the Registrar has validated the enrollment.
 - 2.4 The system displays the checklist of submitted and missing documents.

Trainer

- 1. The Trainers are responsible for the academic evaluation of students. In order to access their class lists, they must first register an account and await administrative approval.
 - 1.1 The system allows the Trainer to request account creation by providing their full name and employee details.
 - 1.2 The system restricts the Trainer from accessing any features until the Registrar approves the account.
 - 1.3 The system shall allow the Trainer to view the subjects assigned to them—viewAssignedSubjects.
 - 1.4 The system allows the Trainer to view the sections assigned to them—viewAssignedClasses.
- 2. To eliminate the need for manual grade sheets, the Trainer must input student grades directly into the system for their assigned subjects.
 - 2.1 The system displays the list of students enrolled in the Trainer's assigned batch and subject.
 - 2.2 The system allows the Trainer to input numerical grades for each student.

2.3 The system prevents the Trainer from inputting invalid characters—letters or symbols—in the grade fields.

2.4 The system allows the Trainer to "Lock" or "Finalize" grades, preventing further edits after submission.

2.5 The system shall allow the Trainer to update grades and remarks for a subject—updateGrade, updateRemarks—via the Subject interface.

3.2.10 Non-Functional Requirements

The non-functional requirements describe the requirements of the users and benchmarks for the system's functionalities.

Functional Suitability

- Functional Completeness: the system must allow the Registrar to encode a complete batch of student documents per section, covering all required fields defined in the StudentRecord.
- Functional Correctness: the viewStudentList() function must provide accurate data that matches the database records with 100% precision.
- Functional Appropriateness: the grade encoding interface must be specifically designed for Trainers, allowing only valid grade inputs—preventing grade entry for non-enrolled students.

Performance

- Time Behaviour:
 - The Registrar and Admin shall be able to search for a specific StudentRecord using a Student ID or Name within 5 seconds.
 - The document upload feature must have a response time of under 3 seconds for files smaller than 5MB.
 - The viewStudentList() page must load within 5 seconds.

- Capacity: the system must support at least 50 concurrent users without degradation of service.
- Resource Utilization: the system must run efficiently on standard school workstations without exceeding 6GB of RAM usage during peak operations.
- The Student Records page–viewStudentList–must reflect information from the database without errors when requested.
- Search results for a specific Student or Batch must load within 3 seconds.

Compatibility

- Interoperability: the system must be able to export reports–e.g., Student lists, grade sheets, student permanent records, Transcript of Records–in standard formats (PDF, CSV, Docx) for use in external auditing tools.
- Co-existence: the system must function correctly on workstations that also run standard antivirus and office productivity software without conflict.

Usability

- Operability:
 - The Registrar shall be able to encode a batch of student documents–per section–within 5 minutes.
 - A trained Trainer shall be able to input and update a student’s final grade within 3 clicks.
 - A trained Registrar shall be able to assign a Trainer to a Subject within 5 clicks.
 - A Student shall be able to complete the–inputStudentData–form within 5 minutes of their first login.
- User Error Protection: The system must validate input fields–preventing letters from finalGrade field–to prevent invalid data entry.
- Learnability: new staff members should be able to perform basic tasks after a 1-hour training session.

Reliability

- Availability: the system must be accessible at least 99.9% within a two-hour window, except during scheduled maintenance periods.
- Fault Tolerance: the system must allow users to queue an update—encoding grades—even if the network connection drops momentarily, synchronizing data once the connection is restored.
- Recoverability: in the event of a system failure, the system must be capable of recovering the database to the last consistent state within 1 hour.

Security

- Confidentiality: users must be logged in to view any private StudentRecord or Document.
- Accountability: the system must provide audit logs that allow the Admin to trace specific actions within 7.2 seconds.
 - Deletion of an account, document, record, grades.
 - Edited account, document, record, grades.
 - Creation of accounts, records, grades.
 - Uploading of Documents.
- Authenticity: the session will time out if a user is inactive for 15 minutes to prevent unauthorized access.

Maintainability

- Analyzability: the system must provide logs that allow developers to trace the root cause of an error within 10 seconds.
- Testability: the total system lines of code (LOC) must be composed of at least 5% LOC dedicated to tracking errors to facilitate testing and debugging.

- Modifiability: the system deployment for updates or patches must be completed within 3 weeks to minimize disruption to the school year.

3.3 Project Details

This section covers the specific details of the methodology used by the researchers to create, deploy, and implement the Aniham Student Record Management System.

3.3.1 Project Development Methodology

The researchers used an Agile approach in the development of the project. The Agile methodology is a project management and software development approach that emphasizes iterative progress, collaboration, and adaptability. Unlike traditional linear methodologies, Agile breaks a project into smaller, manageable units known as iterations or sprints, typically lasting two to four weeks. Each iteration delivers a potentially usable product increment, allowing the researchers to gather feedback early and make continuous improvements (Beck et al., 2001). Agile prioritizes collaboration among team members and stakeholders, encouraging frequent communication to ensure that project requirements remain aligned with user needs and organizational goals (Highsmith, 2009). The focus on iterative development fosters a culture of ongoing improvement, where processes and deliverables are regularly evaluated to enhance quality, efficiency, and user satisfaction (Schwaber & Sutherland, 2020).

3.3.1.1 Planning

The researchers conducted interviews with the client—Aniham Administration and Aniham Students—to understand their operations and business process. The team identified the root problem: the inefficiency of physical document management and the overloaded duties of Registrar.

After finding the root problem, the researchers researched possible systems that could help address the client's needs and weighed them using Analysis of Alternatives.

3.3.1.2 Requirements Analysis

To gather the requirements, the researchers conducted meetings and onsite observations of the school's facilities. This process helped identify the necessary data fields for the Student, Subject, and Document classes and the specific access rights required for the Registrar versus the Trainer.

3.3.1.3 System Design

The system design phase focused on establishing the overall architecture, data structures, and user interfaces. Insights gathered revealed a role-based operational structure where the Registrar and the Admin act as the central authority. To support this workflow, the system adopted the Model-View-Controller (MVC) architecture, which is recognized for its clear separation of concerns and its ability to isolate business logic from user interface components (Gamma et al., 1994). This separation enabled the system to enforce role-specific access, ensuring that each staff member interacts only with functionalities aligned with their responsibilities.

3.3.1.4 Coding

Development will take place in different sprints. Requirements and tasks will be based on the user stories. Features will be tested iteratively to ensure that the system meets the needs of the clients.

Coding will make use of Java 25, an object-oriented programming language, for the backend of the system. Development will follow the Springboot 4.0.4, Spring Security 7, Spring Validation, and Spring Data JPA frameworks. T

Coding will make use of Java 25, an object-oriented programming language, for the backend of the system. The frontend will use HTML, CSS, and JavaScript for responsive user interactions. MySQL will serve as the database platform for its scalability, security, and suitability to the client's existing infrastructure. The project will make use of NetBeans as the development environment for the backend and frontend of the system. Additionally, GitHub will be used for collaboration between the developers, version control, issue tracking, and documentation.

3.3.1.5 Testing

Testing will be conducted throughout the Agile sprints. Functional testing will verify that core features like `login()`, `updateGrade()`, and `uploadDocument()` perform correctly according to specifications. Usability testing will then be carried out using structured task scenarios, where participants representing actual users—Trainers, Students—perform system-related activities under observation.

3.3.1.6 Deployment

The Aniham Student Record Management System will be deployed locally on a secure in-house server within the school premises to ensure centralized data control. Unlike cloud-based solutions, this system is configured for exclusive access through the institution's local network infrastructure. The primary administrative module, which grants full control over student records and document management, is strictly restricted to the Registrar's workstation. However, the system is designed as a web-based application to accommodate other stakeholders via school desktop computers or approved personal laptops. Trainers may access the grading portal by registering an account subject to the Registrar's approval, or through credentials pre-created by the Admin. Students are required to register and create an account before gaining access to the enrollment module; once registered, they

may input their personal information using either designated school terminals or their own personal laptops/desktops. To facilitate a smooth transition, the Registrar, Faculty, and School Director will undergo an orientation regarding the system's use cases and security protocols. A comprehensive user manual will be developed and distributed, providing step-by-step instructions for key functions—inputStudentData for students and viewAssignedSubjects for trainers. Finally, the system will be strictly limited to desktop and laptop environments, with no mobile accessibility enabled, to maintain security and consistent user interface standards.

3.3.1.7 Maintenance

After deployment, the system will undergo continuous monitoring to identify and resolve technical issues. The researchers will provide ongoing support to the school staff for system updates and troubleshooting.

3.3.2 Evaluation Procedure

The evaluation of the Student Record Management System will be done through multiple phases: software evaluation, user acceptance testing, time and motion study, and KPI assessment. First, IT experts will assess the system using an ISO/IEC 25010 to examine the functionality, performance, compatibility, usability, reliability, security, and maintainability of the system. Next, the users involved, mainly the registrar, will undergo User Acceptance Testing. The registrar will perform tasks using the system, and their completion time, error rate, success rate, and feedback will be recorded. To evaluate the system against the original method, a time and motion study will be conducted to compare their task duration, number of steps, and errors. Lastly, the KPIs from the objectives will be

evaluated, including improvements in processing efficiency, document preparation time, data accuracy, user accessibility, and data security.

3.3.3 Software Quality Characteristics

1. Functional Stability

Functional stability refers to how well the system provides the essential features needed for managing student records and generating TESDA Special Order documents. This includes the functions of record encoding, document generation, and data verification, all of which need to perform correctly and consistently. Functionality ensures that the system fulfills its main purpose of supporting the registrar's task through automated and consistent processes.

2. Performance Efficiency

Performance efficiency refers to how well the system performs its task relative to the time and resources it uses. This includes response time, processing speed, and the ability to handle multiple records or requests without any delays. An efficient system would allow the registrar to retrieve and update records quickly, generate TESDA Special Orders faster, and switch between tasks quickly.

3. Compatibility

Compatibility assess how well the system coexists with other software or tools. This includes the ability to export reports in standard formats and operate alongside commonly used applications, including antivirus software and office productivity software.

4. Usability

Usability measures how easy the system is to learn, understand, and operate for the users, especially for the registrar, who will be the main user of the system. A usable student records management system would include a clean and intuitive interface. A logically organized workflow of menus and forms. It also means minimized complexity, error recovery, and clear feedback. This is an essential characteristic because it directly affects the user satisfaction and productivity of the users, mainly the registrar.

5. Reliability

Reliability refers to how stable and consistent the student records management system will be during everyday operations. The system must be capable of functioning even in cases such as network interruptions, simultaneous user activity, and or large data retrieval operations. Reliability ensures that the student's documents and information will remain accessible, intact, and secure.

6. Security

Security measures how well the system protects confidential student information and prevents unauthorized access. The SRMS incorporates secure login, role-based access, and data validation to ensure only authorized personnel can access or modify records.

7. Maintainability

Maintainability evaluates how easily the system can be updated, debugged, and improved overtime. This includes clarity of system logs, identification and resolving of issues, and ability to modify the system without causing major disruptions. This ensures long-term stability and sustainability.

3.3.4 Data Gathering Procedure

To gather the necessary data and requirements for the design of the Aniham Student Record Management System, the researchers focused on extracting first-hand insights from the primary stakeholders through targeted inquiry. First, in-depth interviews were conducted with the Registrar to gain a deep understanding of the existing administrative workflows. These sessions specifically targeted the technical details of how the process of encoding and retrieving student records and documents is currently handled, as well as the operational limitations of their current system. A critical part of this gathering process involved investigating the delays associated with the issuance of Special Order (SO) numbers from TESDA, which are a mandatory requirement for graduating students to receive their certificates and diplomas. Second, data was gathered from the Students to understand the end-user perspective. The researchers inquired about their personal experiences with the current record management setup, with a specific focus on how their enrollment process went. This dual approach ensured that the system design would effectively address the complex compliance needs of the Registrar while simultaneously streamlining the intake process for the Students.

3.3.5 Population Sampling

The population sample consists of Registrar, Human Resource (HR), Students of the institution. Using purposive sampling, the researchers specifically selected participants whose roles required engagement with student records, ensuring that the data collected would be relevant and experience-based (Etikan et al., 2016).

3.3.6 Research Instruments

The primary research instruments employed to collect data were the Interview Guide and the Survey Questionnaire, tailored to the distinct roles of the respondents. The Interview Guide used for the Registrar was composed of specific questions designed to investigate the intricacies of the encoding process, the retrieval of student records, and the functionality of the current system. Crucially, this guide included specific questions regarding external dependencies, such as the causes and impacts of delays in securing TESDA Special Order numbers for graduates. For the Students, a Survey Questionnaire was utilized to gather quantitative and qualitative data regarding their experiences. This instrument focused on evaluating the efficiency of the enrollment process and identifying common pain points in their interaction with school records. Additionally, the System Usability Scale (SUS), adopted from Brooke (1996), will be used during the evaluation phase to measure the overall usability of the proposed system once developed.

3.3.7 Statistical Treatment of Data

Using the System Usability Scale (SUS) of Brooke (1996), user satisfaction will be measured using a 10-question survey that can be answered on a scale of 1-5, 1 being strongly disagree and 5 being strongly agree. The list of questions and scaling of the SUS are presented in Table 3.3.7.1

Table 3.3.7.1 SUS Questionnaire and Likert Scale

#	Statement	1	2	3	4	5
1	I think that I would like to use this system frequently.	☐	☐	☐	☐	☐

2	I found the system unnecessarily complex.	☐	☐	☐	☐	☐
3	I thought the system was easy to use.	☐	☐	☐	☐	☐
4	I think that I would need the support of a technical person to be able to use this system.	☐	☐	☐	☐	☐
5	I found the various functions in this system were well integrated.	☐	☐	☐	☐	☐
6	I thought there was too much inconsistency in this system.	☐	☐	☐	☐	☐
7	I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☐
8	I found the system very cumbersome to use.	☐	☐	☐	☐	☐
9	I felt very confident using the system.	☐	☐	☐	☐	☐
10	I needed to learn a lot of things before I could get going with this system.	☐	☐	☐	☐	☐

The SUS score is calculated by getting the sum of the score contributions from each item. Each item's score contribution differs from a range of 0 to 4. For items 1,3,5,7, and 9, the score

contribution is (the scale position - 1), while for items 2,3,6,8, and 10, the contribution is (5 - the scale position). The sum is then multiplied by 2.5 to get the overall SUS value.

$$SUS\ value = \Sigma(Scores) \times 2.5$$

In addition to the calculated SUS value, the collected data from the SUS will also be treated using frequency, percentage, and weighted mean. These will show the user satisfaction and to determine if the selected ISO 25010 qualities have been met in the SRMS. The frequency and percentage will show the distribution of responses, while the weighted mean will show the overall user perception of the quality of the system. The weighted mean will be calculated by the product of the SUS Likert scale weight and the response frequency, divided by the total number of samples. This formula is as follows:

$$Weighted\ Mean = \frac{(w \times f)}{n}$$

Where:

- w = weight of each SUS likert scale value
- f = frequency of each responses
- n = respondent sample size

3.3.8 Project Cost

The estimated cost of the project will reflect the necessary hardware and software requirements for the development of the proposed SRMS. Table 3.3.8.1 illustrates the breakdown of the costs.

Table 3.3.8.1 Project Cost

Item	Estimated Cost (PHP)
Software Tools (IDE, DBMS, GitHub)	PHP 0.00
Dedicated Local Server Unit	PHP 35,000.00
Uninterruptible Power Supply (UPS)	PHP 6,000.00
External Backup Drive	PHP 4,000.00
Printing and Documentation Materials	PHP 1,500.00
Development Costs (for 3 Developers)	PHP 150,000.00
Total Estimated Cost	PHP 196,500.00

The total estimated cost of the project will be PHP 196,500.00. The responsibility of the costs for the dedicated local server unit, UPS, and external backup drive will be handled by Aniham Technical School, while the printing and documentation materials will be shouldered by the researchers of this study.

3.3.9 Project Risk Management Plan

The project is expected to have risks, and a risk management plan is necessary to resolve or overcome these risks. Table 3.3.9.1 is a 3x3 risk matrix of the possible risks that the project may encounter, while Table 3.3.9.2 shows a detailed breakdown of the risks, risk priority, and risk response.

Table 3.3.9.1 Risk Matrix of possible risks

Likelihood\Impact	Low Impact (1)	Medium Impact (2)	High Impact (3)
Unlikely (1)	Resignation of Trainers Client asks for more features	Database connection issues Data loss or corruption Time management issues User adoption risk	User dissatisfaction Technical incompatibility
Likely (2)	Over budget Client becomes unreachable		Resignation of the Registrar Insufficient funds for the project
Highly Likely (3)			Transportation issues

Table 3.3.9.2 Risk Table

Risk Type	Risk	Impact	Risk Priority	Risk Response
Financial	Over budget	Client will have to shoulder more costs	Low priority	Evaluate alternatives for resources
External	Resignation of the Registrar	Key contact and primary user of the system will be lost	High priority	Train other staff how to operate the system
External	Resignation of Trainers	Secondary users for the system will be lost	Low priority	Train other staff how to operate the system
Strategic	User Disatisfaction	The system did not meet the ISO 250170 standard and does not meet the project	Medium priority	Evaluate the needs of the client and are met

		objectives		
External	Client asks for more features	Overloaded features to be developed	Low priority	Evaluate the needs of the client
Operational	Database connection issues	System will not work as intended	High priority	Conduct multiple system testing
Technical	Technical Incompatability	System is not compatible with client's existing infrastructure	Medium priority	Evaluate the client's infrastructure and needs
Operational	Data loss or corruption	Digitized student data will be lost	High priority	Conduct multiple system testing
External	Time management issues	Development of system will stall	Low priority	Set internal deadlines and adjust if necessary
External	Transportation issues	Inability to meet with client and deploy the system	High priority	Evaluate transportation options and schedule visits in advance
External	Client becomes unreachable	Inability to communicate with client	Low priority	Schedule frequent meetings and assign additional contacts
Technical	User adoption risk	Client will not be able to use the system	Low priority	Train staff on how to operate the system through a comprehensive training plan
Financial	Insufficient funds for the project	Client may not fund the project	High priority	Consult with client on a feasible budget

Based on the risk assessment, the risks with the highest priority are the resignation of the registrar, database connection issues, data loss or corruption, transportation issues, and insufficient funds. The other risks are under low to medium priorities. The overall risk response is to have frequent consultations with the client to gather detailed information on their needs to ensure the system to be developed meets them—fulfilling

the project objectives. The researchers also recognize that there may be other potential risks not accounted for and should be ready to adjust the project. Should any of the risk responses fail, the researchers is to take action in remedying an alternative response to prevent or at least mitigate the risk. If not possible, accept the risk and adjust the project accordingly.

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Appendices

Appendix A Minutes of the Meeting (November 19, 2025)

Anihan
- Minutes of the Meeting -

Date:
November 19, 2025

Time:
9:00 - 4:00 PM

Venue:
Anihan Technical School

Purpose of the Meeting:
Observation

Observations in the Office of the Registrar:

Google drive

- Anihan.edu.ph
- Containing 10Tb of storage

Student Records

- Currently, the records from 2009-2025 are the ones being held in the room
- They also have an archive room in another building containing their past records

Files

Grade sheet

- Gets the Special Order #, which is done using a soft copy

Student Permanent Record

- From SHS
- Takes DepEd 6 months to get back to her (idk if this is referring to record or to special order)

Each student's record is being held in a green folder per student. Inside are the following files:

1. Admission form
2. Interview evaluation
3. SHS student's permanent record
4. Good morale certificate
5. Certificate of Indigency
6. Scholarship agreement
 - a. Signed by the director
7. Attendance record
8. Learners report
9. Birth certificate

Digitization

- The current method is to input the details into Excel, and mail-merge the sheet with the Word file containing the template needed

Special orders

- These are needed for students to go abroad
- A month after graduation, a special order
- This needs to be done 3 times per qualification
 - Bread and pastry
 - Cookery
 - Food and beverage
- A special order takes around 8 months to get back to the school
- Currently, there are 3 batches delayed
 - The batches from August 2024 do not have their special order
 - QSRO5
 - QSRO6
 - CARS1

The special order contains the following files:

1. List of students by 20/25
2. Certificate of Technical and Vocational Education Training (TVET) program
3. On-the-job training (OJT) report
4. Student information
 - a. Start and end date
 - b. Employment status
5. Attendance sheet
6. Registry of workers assessed and certified
7. Student's permanent record
8. Birth certificate

Duties of The Registrar (Ms. Melesine Velasco)

- Handle Student Records
- Is a TESDA officer

Enrollment Process

- Students hear about Aniham via:
 - Referral
 - School/Community Campaign
 - Social Media
 - Former Applicant (Word of mouth)
- Students apply for Aniham via:
 - Email (from website)
 - Students will receive a Google Form
 - Google Forms (from Facebook)
 - Onsite

- Information asked in the Google Form:
 - Personal Information
 - Medical Information
 - Educational Background
- Application via online (Email and Google Forms) will still require students to submit physical documents
 - Students will also be required to answer an Index Card that asks for the same information that is in the Google Form
- The student is then interviewed by the Recruitment team (online or onsite)
 - Students can officially enroll, depending on the evaluation

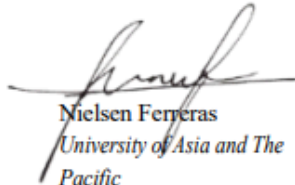
Student Records Management Process

- Registrar receives student documents and information from:
 - Google Forms Google Sheet responses
 - Index cards
- Both Google Sheets responses and Index cards are used to double-check if the student provided the correct information
- Student information is inputted into a central Excel spreadsheet, which contains:
 - Student information
 - Grades
- Documents are stored in green folders (one folder per student), which are stored in the cabinet of the Registrar's office
- Throughout the year, Trainers provide the Registrar with the students' grades via:
 - Printed gradesheet (most preferred)
 - Excel gradesheet
- Registrar updates the students' grades in the main Excel sheet
- From the main spreadsheet, the Registrar inputs the Students' permanent record and Transcript of Records (TOR)

Special Orders Filing and Submission Process

- Registrar prints and compiles the necessary documents for filing a Special Order
- Special orders (SO) must be filed at least one month after students' graduation
- One SO per 20-25 students and per qualification (There are 3 of them under CARS)
 - Assuming there are 80 students per batch, the Registrar would need to file a total of 12 SOs
- Submits the documents to TESDA
- TESDA reviews and clears the students
- **Current Issue:** 3 Batches from 2024 have yet to have their SO filed to TESDA

Compiled by The Researchers:



Nielsen Ferreras
*University of Asia and The
Pacific*
3BSIT Student

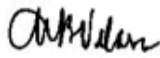


Sean Ligan
*University of Asia and The
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Emmanuel H Velasco
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Acknowledged by:



Ms. Melesine Velasco
Assistant to the Registrar
Anihan Technical School

Appendix B Minutes of the Meeting (February 13, 2026)

Purpose: Project Realignment - Anihan Technical School

Date: February 13, 2026

Time: 2:00 PM to 3:30 PM

Location: Online Zoom Meeting

Attendees:

- Ms. Melesine Velasco (Assistant to the Registrar)
- Mark Ferraras (Student)
- Sean Ligan (Student)
- Emmanuel II Velasco (Student)

Apologies/Regrets:

- Ms. Melesine Velasco - Unable to meet in previous days due to recent priorities within Anihan Technical School

Agenda Review:

1. Presentation of Project Proposal
2. Q&A
3. Plan Moving Forward

Discussion Points:

1. Presentation of Project Proposal:
The group presented the project proposal that was approved after the defense. The presentation covered the entirety of Chapter 1, Assessment of Alternatives, System Analysis, System Design, and Project Cost from the paper.
2. Updates From Anihan:
Ms. Sen informed the group that Anihan has recently applied to become a TESDA assessment center, in addition to being a Training Center. Additionally, the school's internet connection is still slow, but upgrades are underway to improve it. Furthermore, a new staff member has been hired to help in the office of the registrar.
3. Overall Comment:
Ms. Sen affirmed that the project will definitely improve the registrar workflow. She shared her appreciation that the project covers all key problem areas.
4. On System's Access:
Ms. Sen asked if anyone could enroll in the system. Mark clarified that the system will only be accessed within the school's premises—cannot be accessed via public internet.
5. Digitization Plan
Mark requested whether it is possible for students to aid in digitizing the student records as a way for them to practice computer skills. Additionally, he also requested if it is

possible to use the computer lab for both digitization and implementation of the system during the school's enrollment period.

6. Early Deployment:

Ms. Sen requested whether it is possible to have the system ready by June or July in time for the enrollment period of the school. Em explained to her the ideal research project timeline that the group was oriented on during Introduction to IT Research from last semester. He informed her that, despite the targeted full implementation of the system by August, the group could try to have it ready before then, but it is still subject to the advice of the group's research adviser and professor.

7. Digitization of Records:

Mark asked Ms. Sen how many physical student records she has currently in the office. She replied that, on average, there are 150 records per year since 2022.

8. Meeting with Other Key Stakeholders:

Ms. Sen recommended that the group re-present the project to the directors and to the new staff on February 24 to get their input, especially on the other previous points.

Action Items:

- | |
|--|
| <ul style="list-style-type: none">• All Team Members: Revise the paper and Digitization Plan |
|--|

Decisions Made / Agreements:

- Nothing official has been agreed on until the next meeting

Open Issues:

- The project budget has yet to be approved by the school's directors

Next Meeting Date:

- The next online Zoom meeting will be on February 24 from 2:00 to 3:00 PM

Adjournment:

- The meeting was adjourned at 3:30 PM.

Prepared by:

Mark Ferraras, Sean Ligan, Emmanuel II Velasco

University of Asia and The Pacific Students

marknielsen.ferreras@uap.asia, seanmichael.ligan@uap.asia, emmanuel.velasco@uap.asia

Approved by:

Ms. Melesine Velasco Assistant to the Registrar	 Signature
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Appendix C Minutes of the Meeting (February 26, 2026)

Purpose: 2nd Project Realignment - Anihan Technical School

Date: February 26, 2026

Time: 3:30 PM to 4:30 PM

Location: Online Zoom Meeting

Attendees:

- Ms. Herminia F. Gabutina (Registrar)
- Ms. Melesine B. Velasco (Assistant to the Registrar)
- Mark Ferraras (Student)
- Sean Ligan (Student)
- Emmanuel II Velasco (Student)

Apologies/Regrets:

- Ms. Minnie informed the group that Anihan does not have the immediate budget to fund the project

Agenda Review:

1. Presentation of Project Proposal
2. Q&A
3. Budget and alternatives
4. Development Timeline

Discussion Points:

1. Presentation of Project Proposal:
The group presented the project again, covering the same topics from the last meeting but with the addition of the budget breakdown, digitization plan, and development timeline.
2. System access and security:
The group clarified that, since the server will be set up in the registrar's office, the system cannot be accessed publicly—only the registrar and assigned users can access it strictly on the school's premises. Anyone wishing to access the system must connect to the school's internet.
3. Data protection
Ms. Minnie asked, since the 2 workstations are connected to the internet via ethernet cable, how can data be protected? The group stated that the system can only be accessed via authorized accounts that are created by the administrator of the system. This would require login authentication to prevent unauthorized entry. For further security, Mark recommended that the system be set up such that it limits access to specific devices.
4. 75% System Completion by May:

Ms. Minnie asked what to expect as a 75% system completion by May. Em explained that Aniham can expect that the server has been set up and three user roles (Admin, Registrar, and Student) are functioning. For the case of the group, it will also include the research paper which is needed in preparation for the second defense this May.

5. Digitization Scope:

Mark clarified that the project will only digitize 150 student records from the previous school year, as encoding every single record would take approximately 2 years. The system will also need to be designed to handle 10-15 students/users accessing the system without it crashing.

6. Lack of Budget Allocation:

Ms. Minnie explained that, since Aniham receives funding from donations, the organization does not have the immediate funds for the project. Additionally, the school just recently started a fundraiser in preparation for the school's 40th anniversary, further limiting the available funds. The group offered the second alternative of using a cloud solution. Mark explained that subscription to the service may cost up to 20K annually. The group also offered a third alternative, which is to use the existing workstations and equipment that the school has. However, the group explained that the system may be limited by the specs of the available equipment. The two registrar staff prefer this option and ask if it is possible to get discounted deals for the equipment directly from suppliers. The group agreed that it is possible, but reiterated these equipment will most likely be second-hand. Currently, there is no set minimum or maximum budget provided by Aniham for the project.

7. On Cloud alternative:

When offered the cloud solution, the registrars declined, stating security reasons and a lack of knowledge or dedicated knowledgeable staff on using the technology. Mark explained that the cloud is safe and recommended that at the next meeting, the group can present some credible and reliable cloud services to convince them. Otherwise, the group will still maintain the third alternative.

8. On subscription/copyright use of the system:

The group mentioned that the system will be copyrighted by the UA&P. Ms. Minnie asked if Aniham has to pay a subscription fee to continue using it, as the school may decide not to use the system later on. The group clarified that there is no subscription fee, but it is a contract that will be signed by both Aniham and UA&P, which states that Aniham cannot distribute or share the system to anyone else. The group also regretted to inform that they do not know the other exact details of the copyright contract, but explained that is what they have been told. The group recommended reaching out to the UA&P for further details.

Action Items:

- | |
|--|
| <ul style="list-style-type: none">• All Team Members: Research credible and reliable cloud solutions, plan out next meetings (both online and onsite), and look for cheaper equipment or deals |
|--|

Decisions Made / Agreements:

- The project is greenlit

Open Issues:

- The group cannot proceed properly without the budget

Next Meeting Date:

- The next online Zoom meeting will either be on March 5, 3:30 - 4:30 PM or March 6 3:00 - 4:00 PM
- The next on-site meeting will be on March 13. This is subject to change.

Adjournment:

- The meeting was adjourned at 4:40 PM.

Prepared by:

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Approved by:

Ms. Herminia F. Gabutina <i>Registrar</i>	Signature 
Ms. Melesine B. Velasco <i>Assistant to the Registrar</i>	Signature 